

Aerodynamics Subsystem

Project title:
Rear Wing Design

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1. Initial Rear Wing Design Concept

1.1. Rules relevant to the Rear wing design (FS rules 2026)

T 2 GENERAL DESIGN REQUIREMENTS

T2.1 Vehicle Configuration

T2.1.3 Open wheel vehicles must satisfy the following, see also figure 4:

- The wheel/tire assembly must be unobstructed when viewed from the side.
- No part of the vehicle may enter a keep-out-zone defined by two lines extending vertically from positions 75 mm in front of and 75 mm behind the outer diameter of the front and rear tires in the the

side view of the vehicle, with steering system straight ahead. This keep-out zone extends laterally from the outside plane of the wheel/tire to the inboard plane of the wheel/tire assembly.

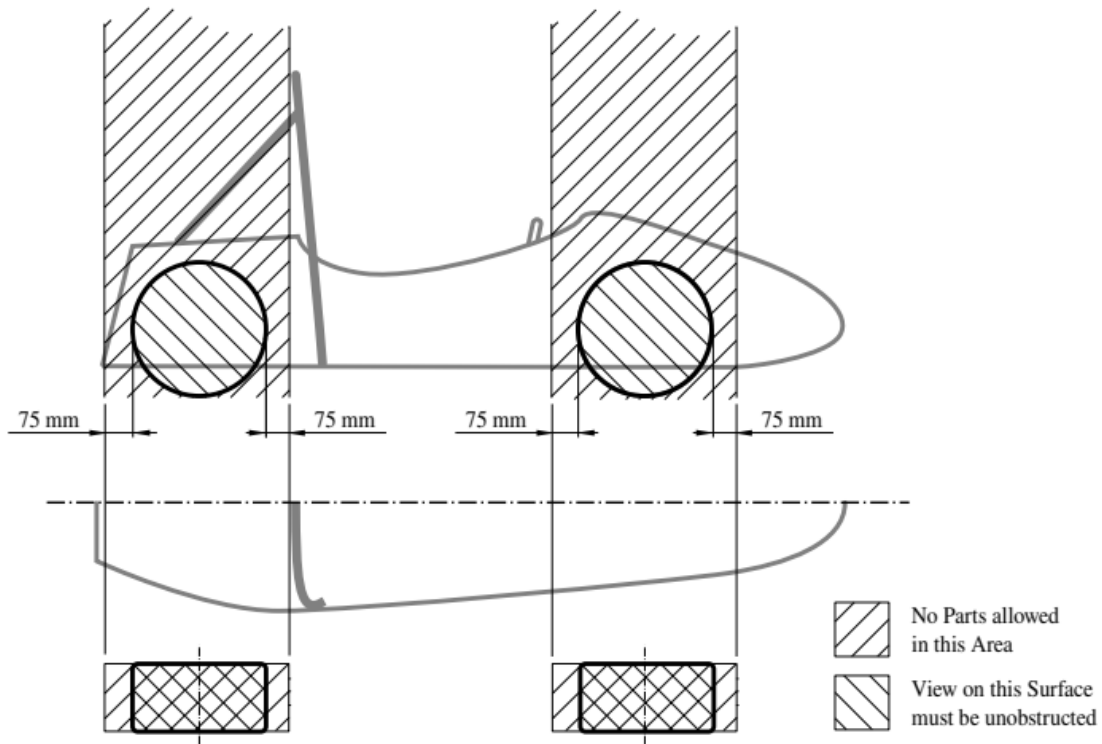


Figure 4: Keep-out-zones for the definition of an open-wheeled vehicle.

T8.2 Restrictions for Aerodynamic Devices

T8.2.1 Height restrictions:

- All aerodynamic devices rearward of a vertical plane through the rearmost portion of the front face of the driver head restraint support, excluding any padding, set to its most rearward position must be lower than 1.1 m from the ground.

T8.2.2 Width restrictions:

- All aerodynamic devices higher than 500 mm from the ground, must not extend outboard of the most inboard point of the rear wheel/tire.

T8.2.3 Length restrictions:

- All aerodynamic devices must not extend further rearward than 250 mm from the rearmost part of the rear tires.

T8.2.4 All restrictions must be fulfilled with the wheels pointing straight and with any suspension setup with or without a driver seated in the vehicle.

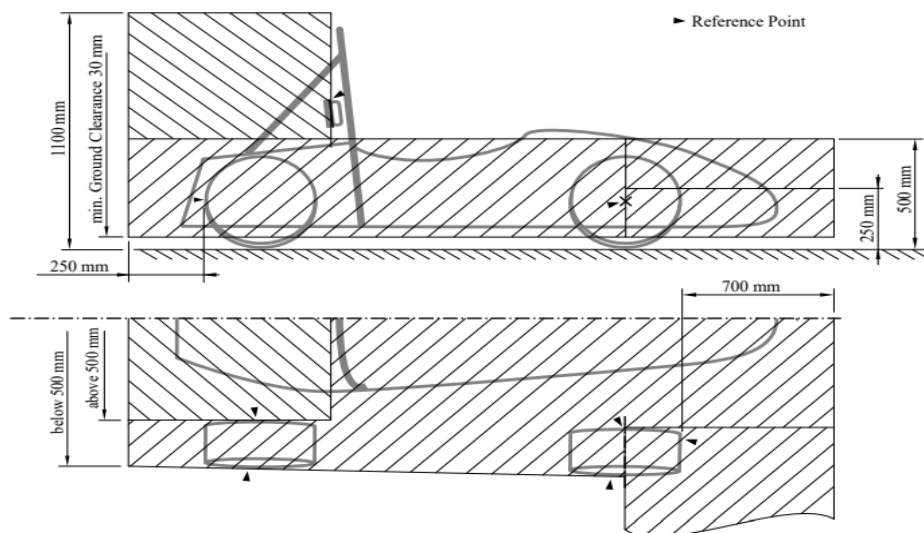


Figure 14: Maximum dimensions and positioning of aerodynamic devices. The positioning space is further restricted, see T2.1.

1.2. Benchmark Study

The detailed benchmark analysis of rear wing designs of FS teams.

<https://docs.google.com/spreadsheets/d/1sd-uz6C1eHQekaGIy17PYtOVVyWpDRJG0WEdknzaJXI/edit?usp=sharing>

1.2.1. Special features of the rear wing elements observed

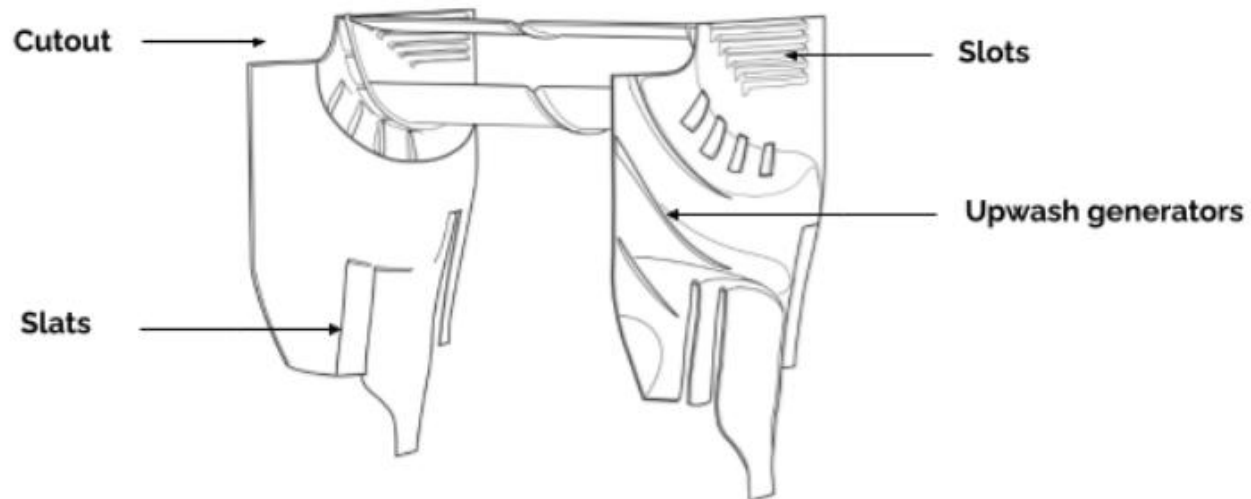
2. Wing elements consist of a varying camber

- Using a wing element with varying camber (often seen as a "dip" or "swan-neck" profile in the center).
- The hoop and the driver's helmet create a shadow of turbulent air directly behind them.
- By reducing the camber or angle of attack in the center, teams prevent the wing from stalling.
- They increase the camber in the outboard sections where the airflow is cleaner and can support higher aerodynamic loads.

3. Extra wing element set utilizing the front space between the endplates.

- An extra element at the front of the endplate acts like a leading-edge slat on an airplane wing. It captures this upward-moving air and re-aligns it, so it hits the main wing elements at a more favorable effective angle of attack.

1.2.2. Special features of the rear wing end plates observed



1. Louvres (Slots)

- **Bleeding high-pressure air**

Louvres are horizontal cuts that allow high-pressure air from the inside of the wing to gradually bleed into the lower-pressure ambient air outside the endplate.

- **Reducing induced drag**

By letting the air bleed out gradually rather than rolling off suddenly, louvres weaken the wingtip vortex and decrease the pressure differential, which directly prevents and reduces induced drag.

- **Strengthening the underwing vortex**

The airflow passing through the louvres enhances the underwing vortex, which increases the overall lift-to-drag coefficient and provides more downforce per unit of drag

2. Cutouts (Top Edge)

- **Targeted airflow improvement**

To precisely manage and improve the airflow in upward direction, thereby increasing the overall aerodynamic efficiency of the car.

- **Weakening wingtip vortices**

The specific shape of the cutout's trailing edge creates a clockwise vortex that deflects and weakens the counter-rotating wingtip vortex, effectively reducing induced drag and manipulating the wake's path.

- **Preventing aerodynamic stall**

Cutouts allow neutral pressure to flow from the outside of the endplate to flow back inside, mixing with and optimizing the low-pressure underwing vortex. This increases the flow velocity downstream, keeping the airflow firmly attached to the wing and reducing the likelihood of a stall

3. Slats (Bottom Edge)

- **Expanding the diffuser flow**

Positioned at the rear bottom of the endplate, slats direct airflow outward to help expand the air exiting the car's diffuser. This expansion promotes an creates a low-pressure wake.

- **Generating more downforce**

Because of the low-pressure wake created by the slats, the diffuser can draw airflow from under the floor much more efficiently, significantly increasing downforce from both the floor and the diffuser.

- **Increasing effective ratio**

By allowing airflow to move from one side of the slat to the other, slats increase the efficiency and the effective aspect ratio of the endplate, which positively impacts the wing profiles between them.

- **Managing wheel wake**

Slats can be splayed outward to act like a perforated Gurney flap, taking expelled airflow and pushing it outwards into the turbulent region behind the rear wheels

4. Upwash Generators (Strakes)

- **Redirecting turbulent air**

Upwash generators are cambered strips placed on the outer surface of the endplate that catch and direct turbulent airflow upwards, managing air that would be obstructed by the endplates.

- **Improving wing efficiency**

They adjust existing airflow patterns over the endplate and bent upwards to boundary layer flow to mix into the vortex.

- **Enhancing rear wing-diffuser interaction**

By supporting upwash flows, these generators improve the aerodynamic interaction between the rear wing and the diffuser.

5. Extra small wing element being connected to lower rear of the end plate from outside.

6. Extruded/Protruded parts

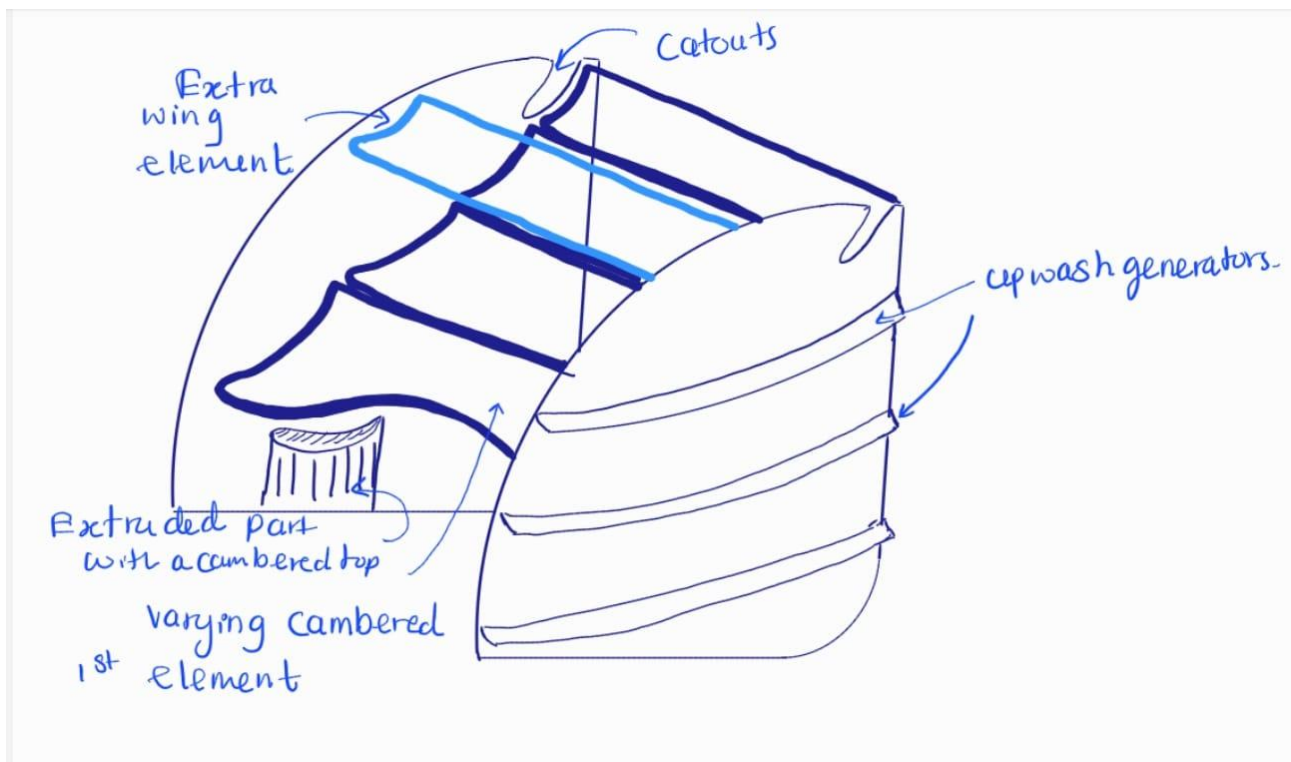
- On inner surface of the end plates under the first element there is a small part protruded out from the inner surface.
- Its top surface has a camber that govern the airflow up.
- The free space at lower part of the end plate has been utilized.

1.3. Initial Design Concept

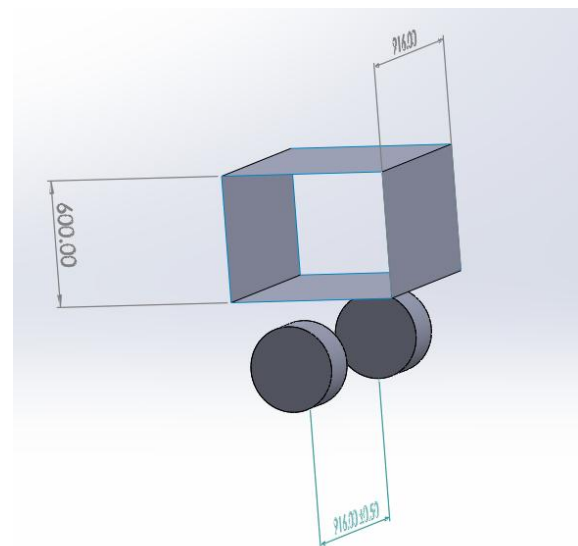
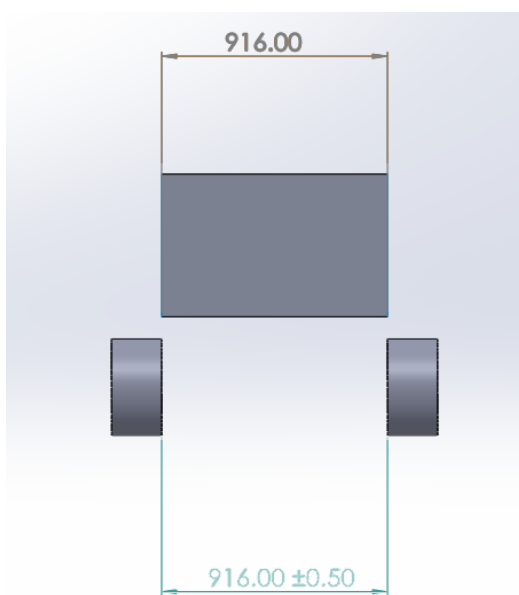
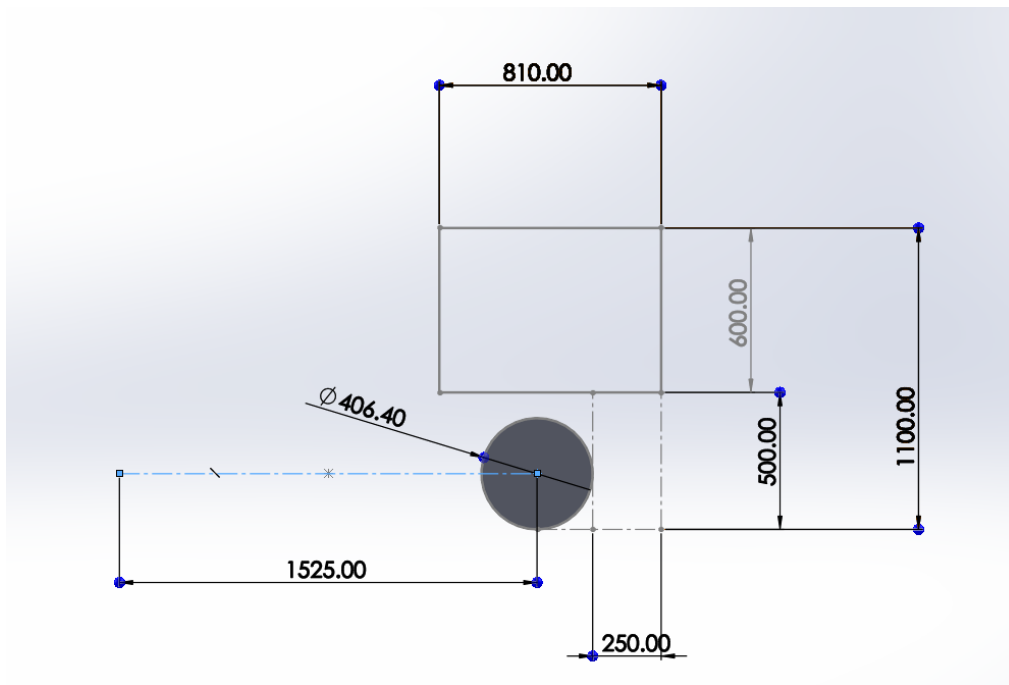
- The rear wing consists of all 5 elements.
- Four elements placed in order of increasing the angle of attack.
- The fifth element is fixed at the front space between the end plates.
- First element to be varying camber

(With the performance and manufacturing possibilities - add the varying camber feature for other elements as well)

- Adding cutouts to the top edge of end plates
- Adding upwash generators on the surface of the endplates
- Small extruded part below first element with cambered top surface.



1.4.Initial limit box dimensions and design targets



The following shows the downforce values generated by aero packages of few formula student teams.

Team	Downforce (N)	Relevant speed (km/h)	CLA	Down force at 55km/h (N)	Source
Tallin	820	55		820	https://www.formulastudent.ee/en/fest24-season/
Monash			5.5	786.3038885	https://www.monashmotorsport.com/m24summary
University of Toronto Formula Racing	1078.7315	110		509.8691855	https://fsaeutoronto.ca/UT25
Formule Polytechnique Montréal	998	80	3.39	471.7109375	https://fsae.polymtl.ca/car
Eforce Parague formula	2941.995	100		889.9534875	https://eforce.cvut.cz/ctu-25/

- Downforce generated by Falcon E Racing E2 – 492N
- The rear wing generates (20 – 30%) from the total downforce of the car.
- As the initial target,

Total downforce = 600N – 700N

Downforce by the rear wing = 150N – 175N

2. Design and Optimization

2.1. Airfoil selection

2.1.1 Airfoils

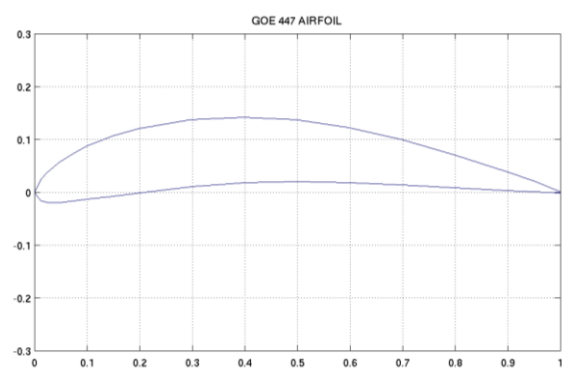
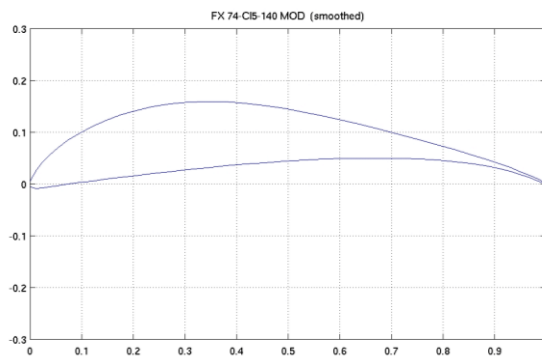
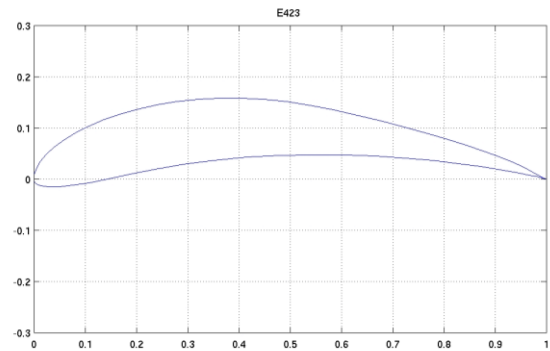
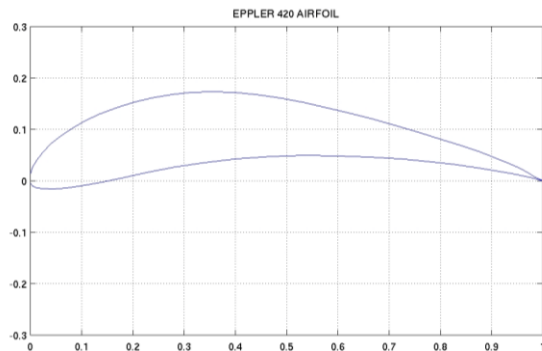
The selected airfoils for further optimizations.

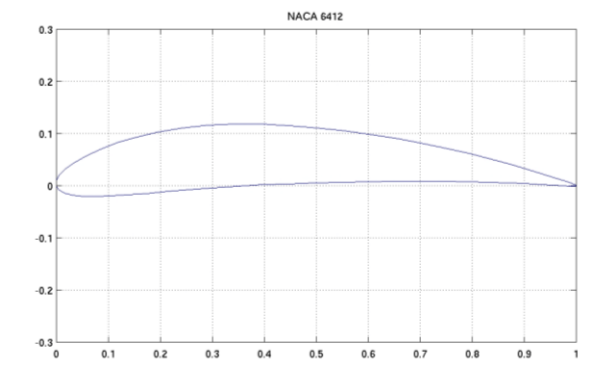
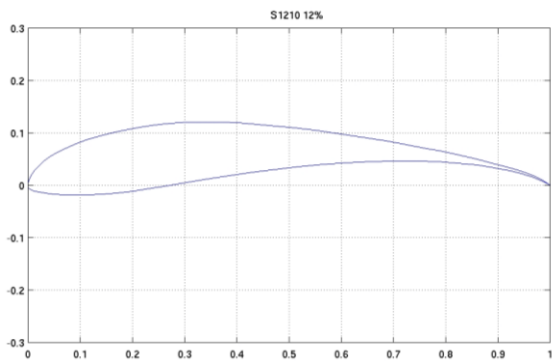
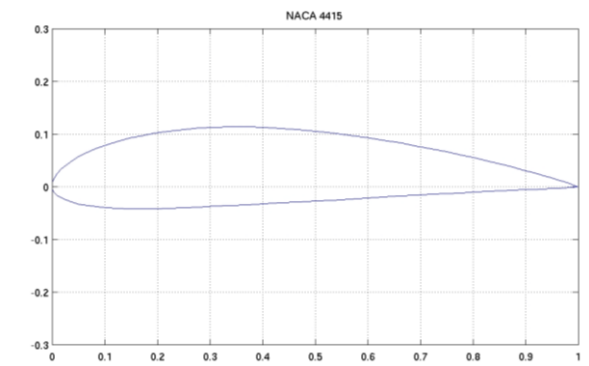
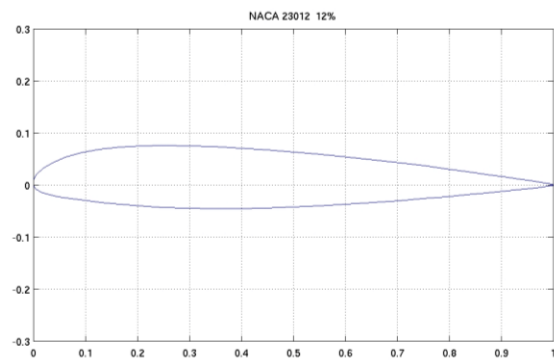
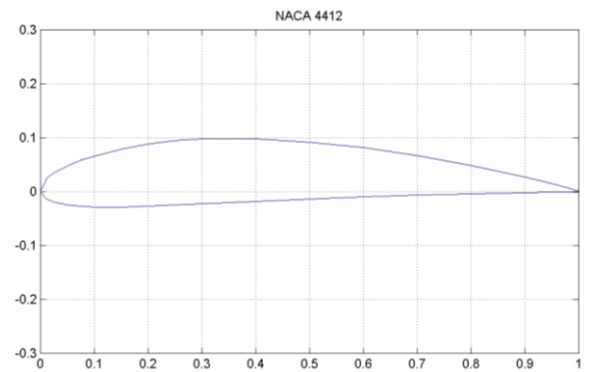
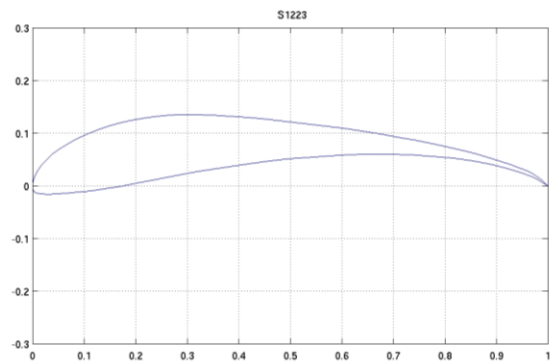
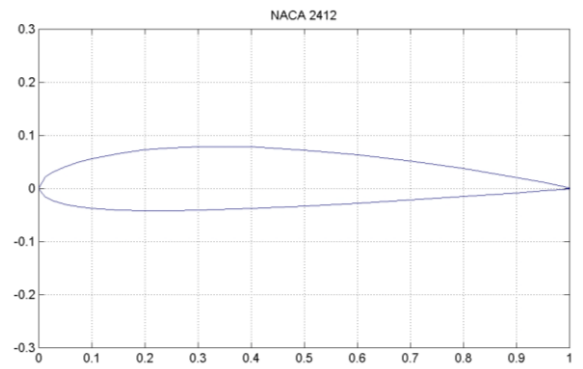
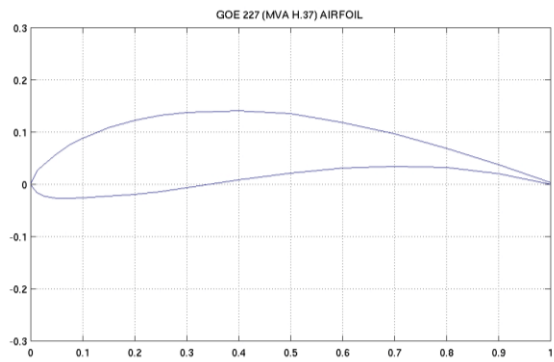
- E420 (Eppler 420)
- E423 (Eppler 423)
- FX 74-CL6-140 (Wortmann FX 74-CL6-140)
- GOE 227 (Göttingen 227)
- GOE 447 (Göttingen 447)
- NACA 2412
- NACA 4412
- NACA 4415
- NACA 6412
- NACA 23012
- S1210 (Selig 1210)
- S1223 (Selig 1223)

Why are those airfoils selected,

- **E423 & S1223:** Selected for their high lift coefficients (CL) specifically designed for low Reynolds number regimes typical of Formula Student speeds.
- **S1210 & E420:** Included to evaluate high-camber profiles that maintain flow attachment at aggressive angles of attack.
- **FX 74-CL6-140:** Chosen for its high thickness-to-chord ratio, providing the necessary internal volume for structural spars and mounting points.
- **NACA 4412 & 4415:** Included as industry-standard high-lift profiles to serve as a reliable performance baseline for comparison against more modern geometries.
- **NACA 2412:** Selected to analyze the trade-off between reduced drag and downforce production in sections of the track where higher speeds are reached.
- **NACA 6412:** Used to study the effects of laminar flow and drag bucket characteristics on the overall efficiency of the wing assembly.

- **NACA 23012:** Included to evaluate an airfoil with a low pitching moment (C_{m0}) to reduce the torsional load and bending moments on the wing mounting struts.
- **GOE 227:** Selected for its unique trailing edge geometry, making it a candidate for a secondary element or flap in a multi-element configuration.
- **GOE 447:** Chosen to test its performance in ground effect and its ability to redirect airflow effectively when positioned behind a main plane.
- **Diverse Camber Range:** The selection provides a wide range of camber percentages to identify the point of flow separation at the expected operating velocities.
- **Pressure Distribution Analysis:** Each airfoil was selected to compare different pressure recovery gradients on the upper surface to prevent premature boundary layer separation.
- **Computational Validation:** The inclusion of well-documented NACA profiles allows for the validation of CFD mesh sensitivity and turbulence model accuracy against experimental wind tunnel data.
- **Manufacturing Feasibility:** The list includes airfoils with varying trailing edge thicknesses to determine the limits of composite layup and manufacturing tolerances for the final wing.



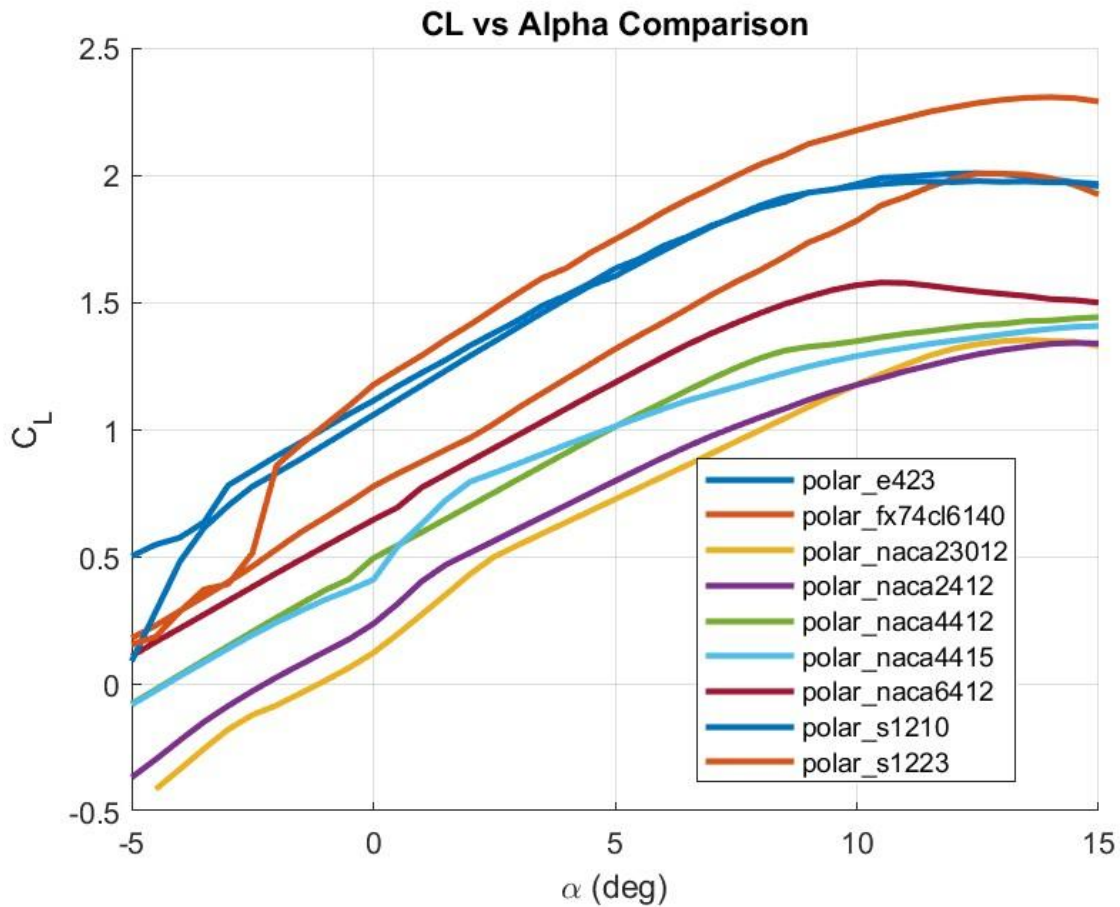


2.1.2. Polar value comparison of airfoils

Velocity – 15 m/s

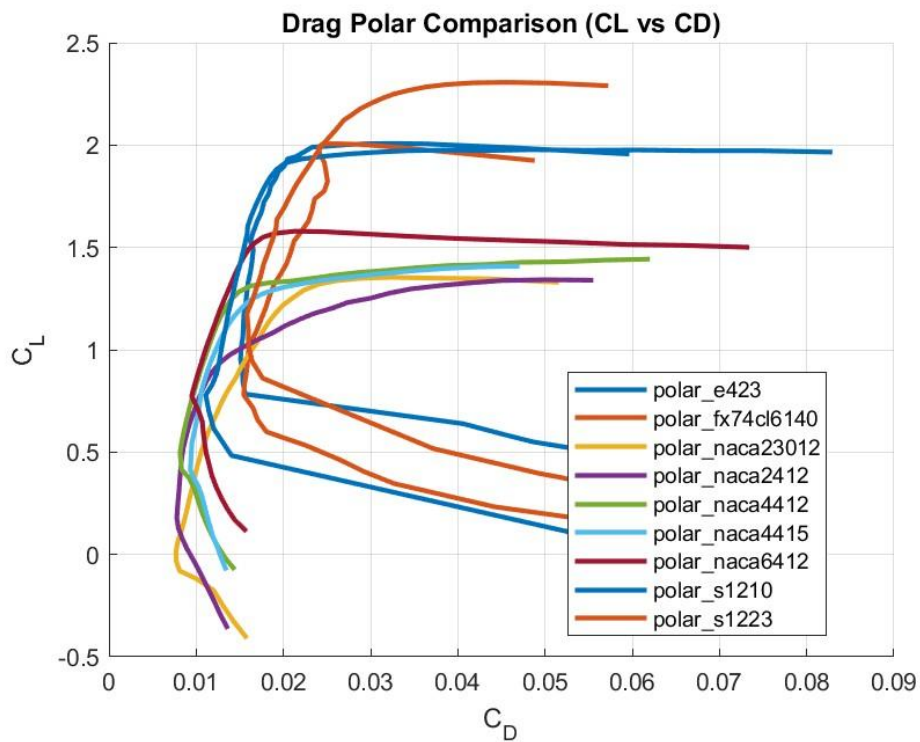
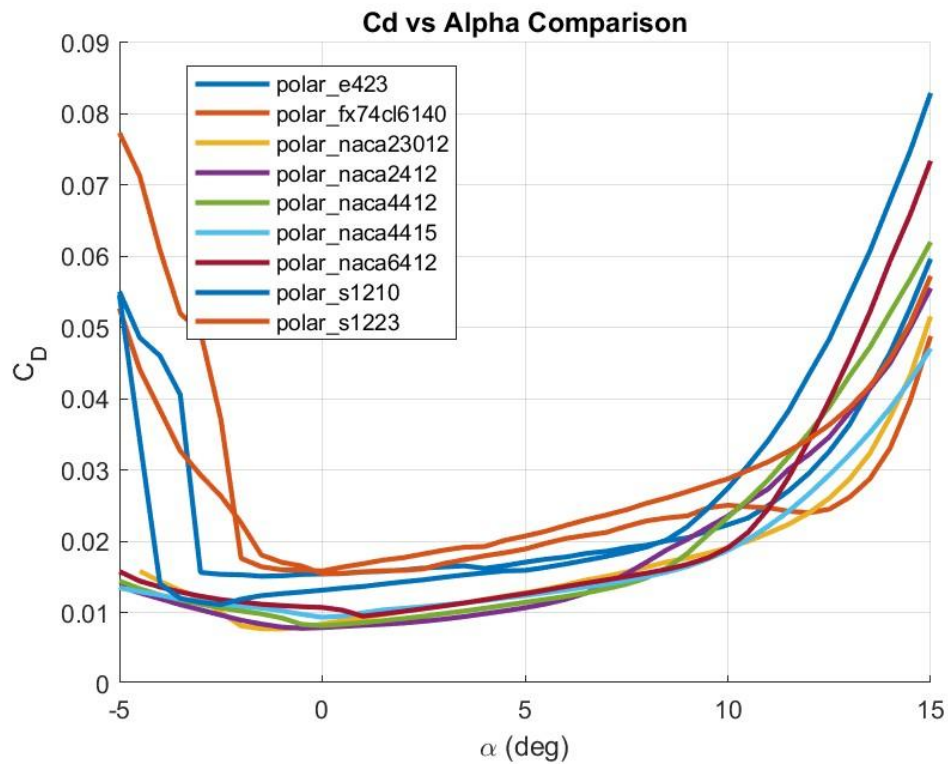
Reynold's number - 0.298×10^6

(Note: the chord length here was considered as 0.3m)



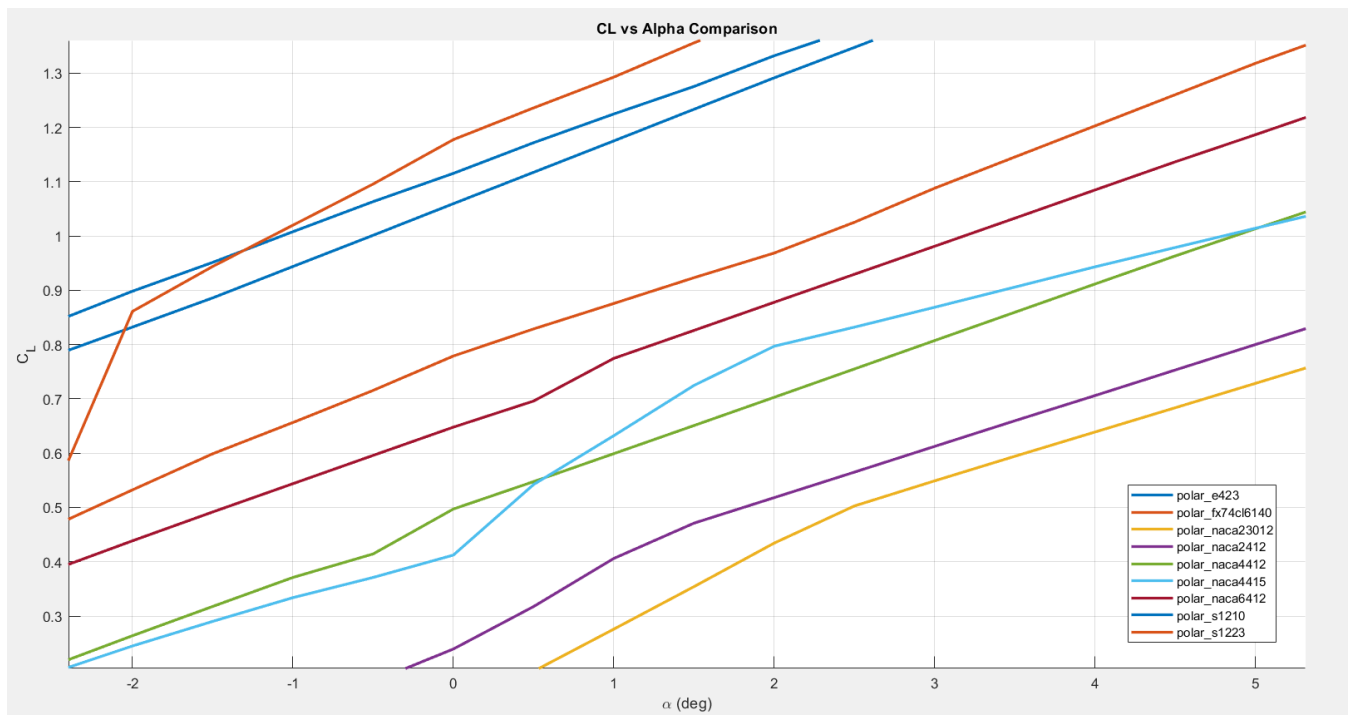
(Note: The coordinate files of G447, G227 and E420 do not respond properly for XFOIL software and Xflr 5 software. So, now those graphs are not shown in this comparison. The coordinates need to be fixed manually)

All the polar values are obtained using the XFOIL software and MATLAB codes are used to run it as well as to generate all the comparison curves.)



CL vs alpha curve for (-2 degrees to 5 degrees)

- High-lift airfoils are optimized for a specific laminar bucket, and focusing on the -2° - 5° range ensures the analysis covers the most efficient operating window where flow remains attached.
- Within this range, the lift-to-drag ratio is typically at its peak, whereas exceeding 5° results in an exponential increase in pressure drag that negatively impacts straight-line acceleration.
- The airflow reaching the rear wing is influenced by vehicle bodywork and upwash, meaning the effective angle of attack is often lower than the physical mounting angle.

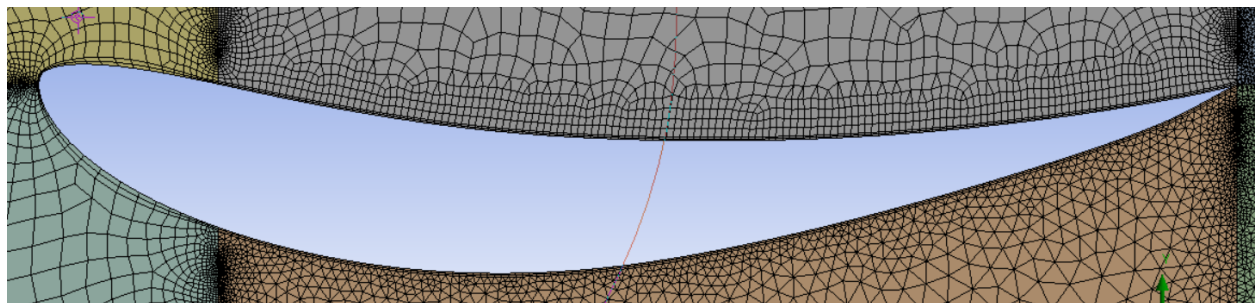
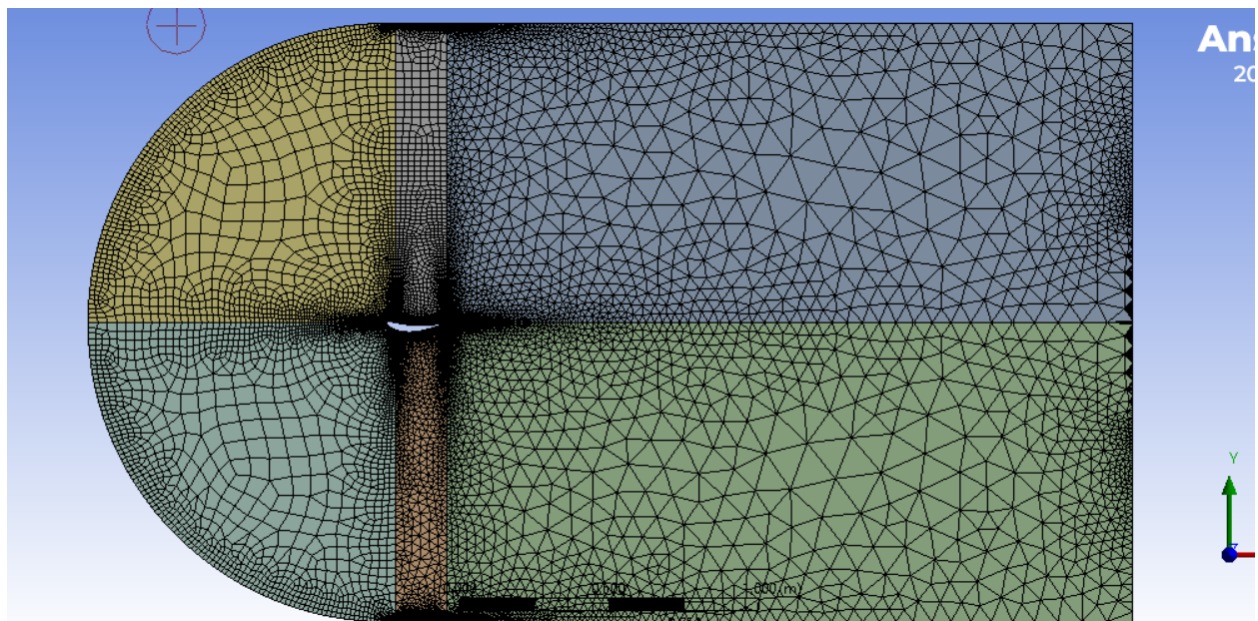
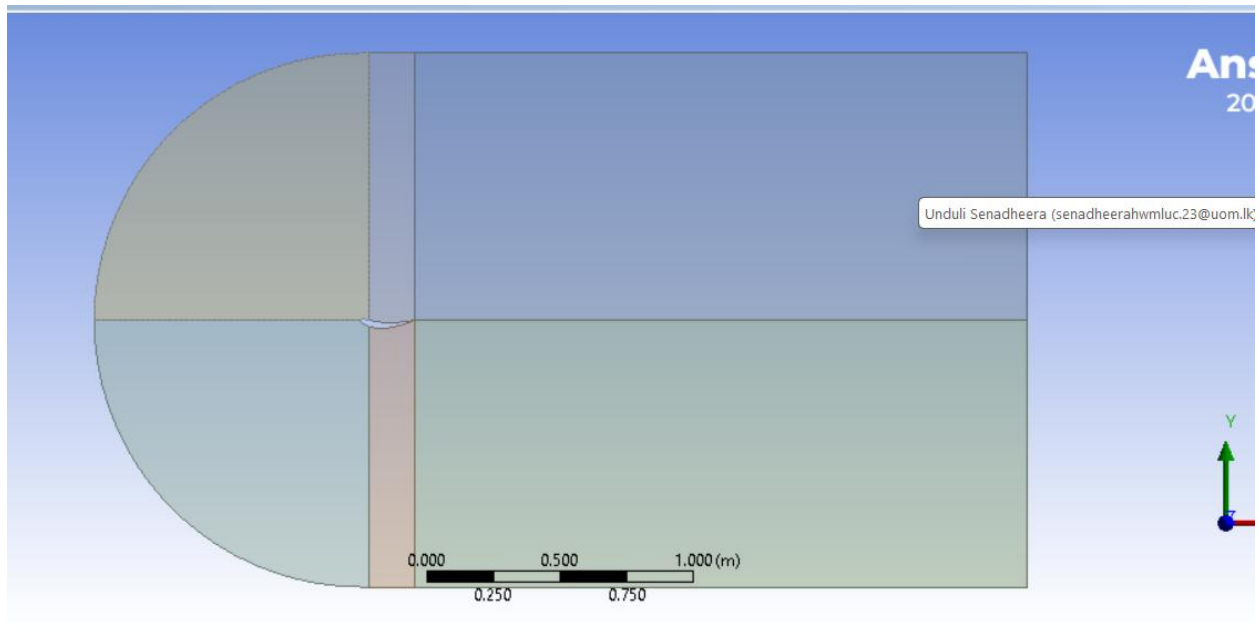


The top leading graphs are,

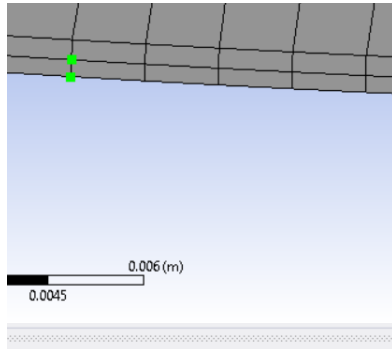
- S1223
- E423
- S1210
- Fx74c16140
- NACA6412

For the Initial stimulations E423 is used.

Flow stimulation for E423

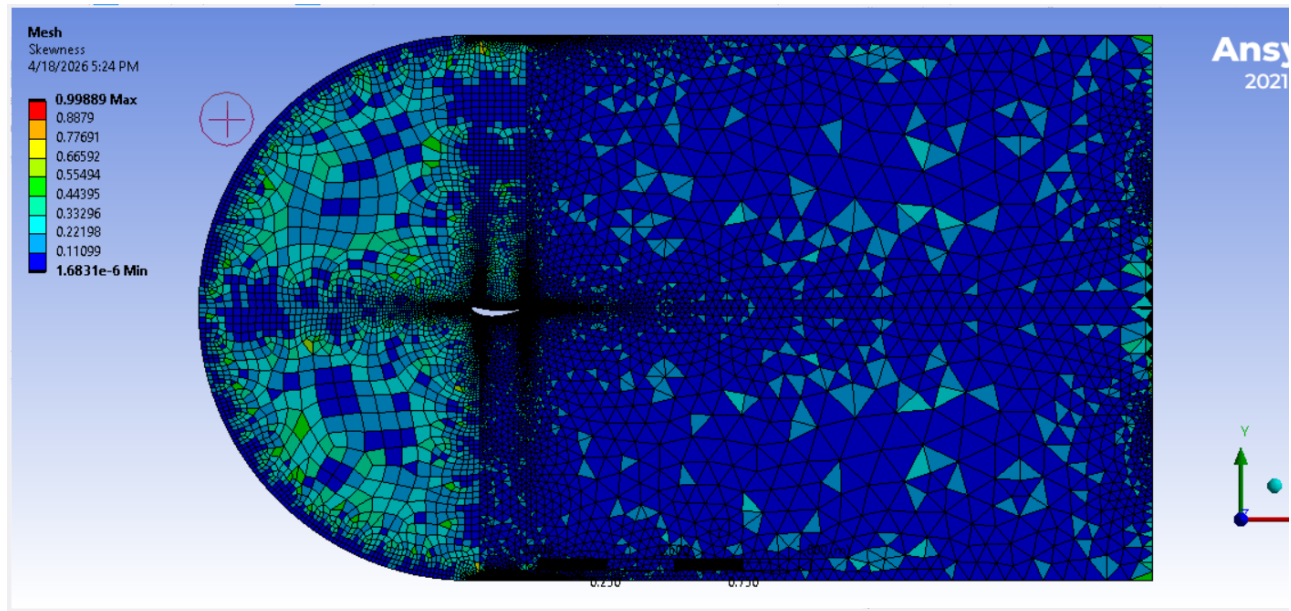


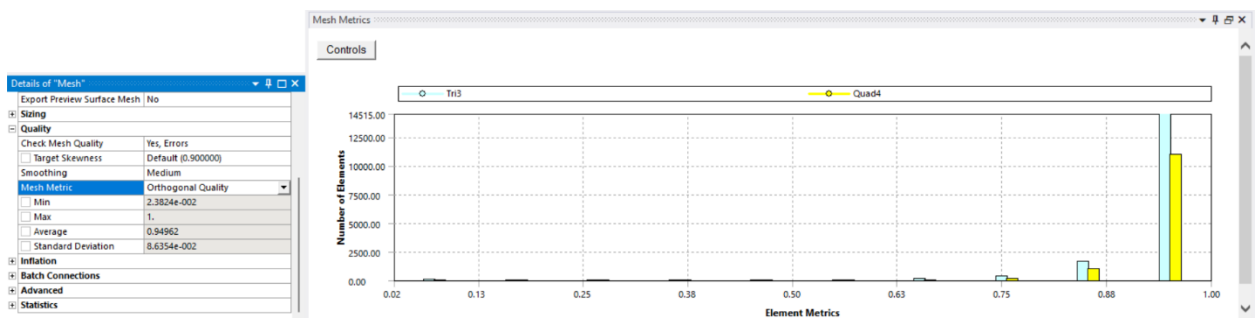
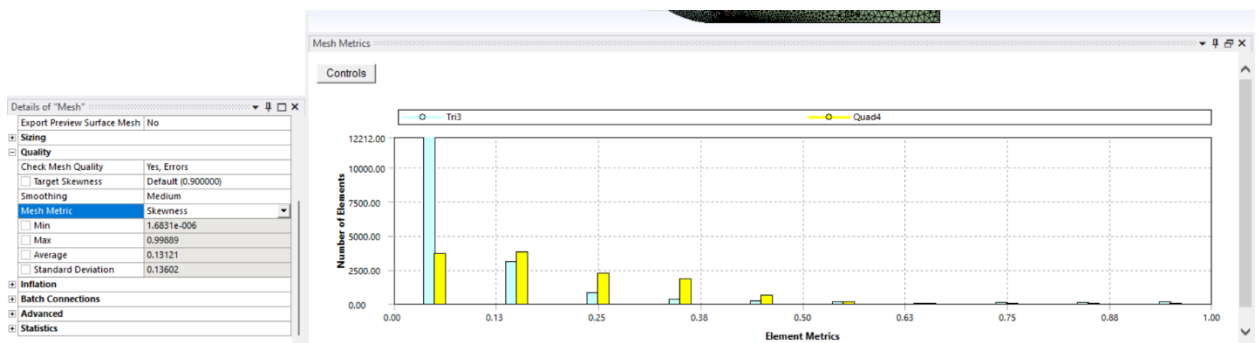
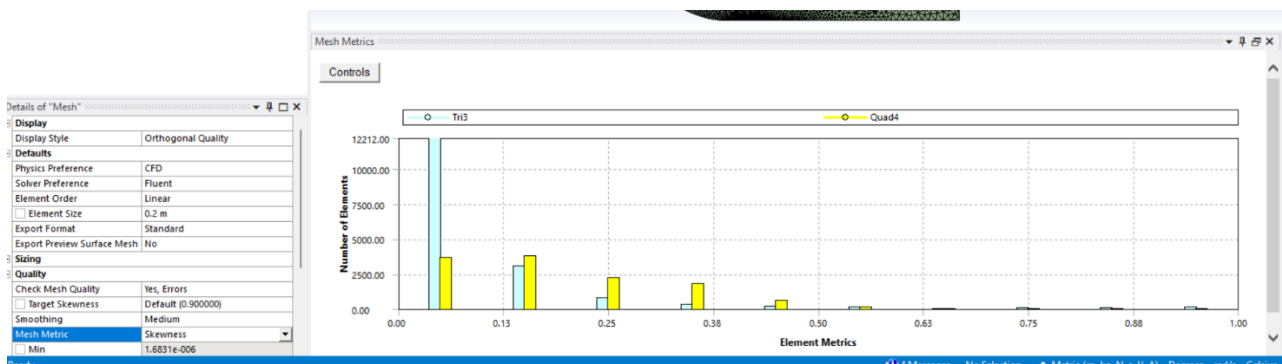
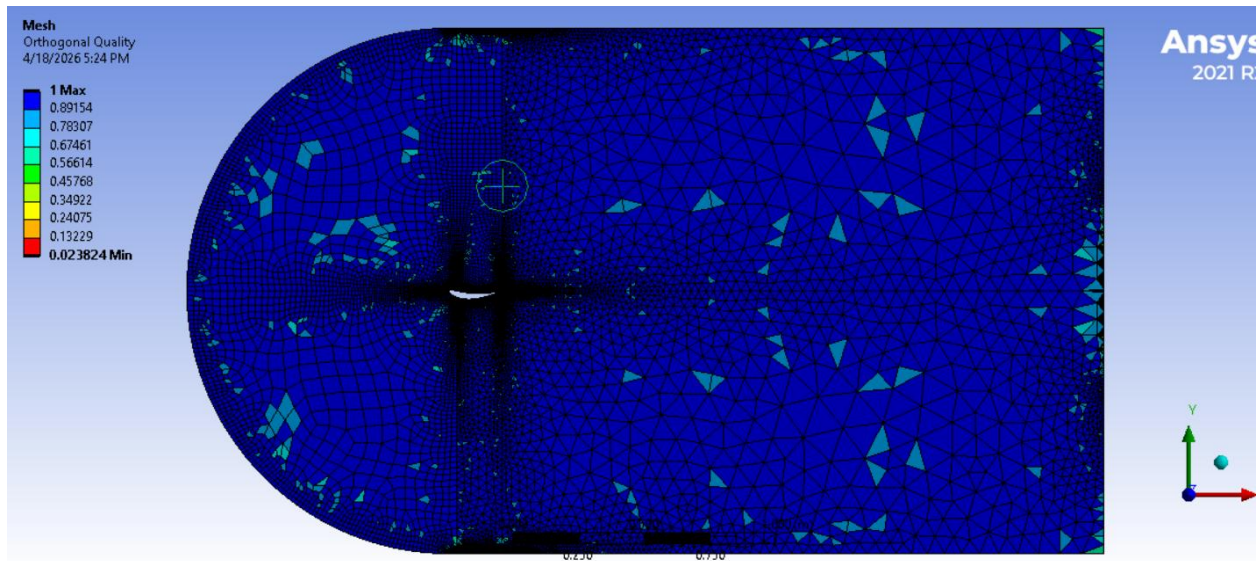
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Geometry	6 Faces
Definition	
Suppressed	No
Boundary Scoping Method	Geometry Selection
Boundary	4 Edges
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Inflation Algorithm	Pre

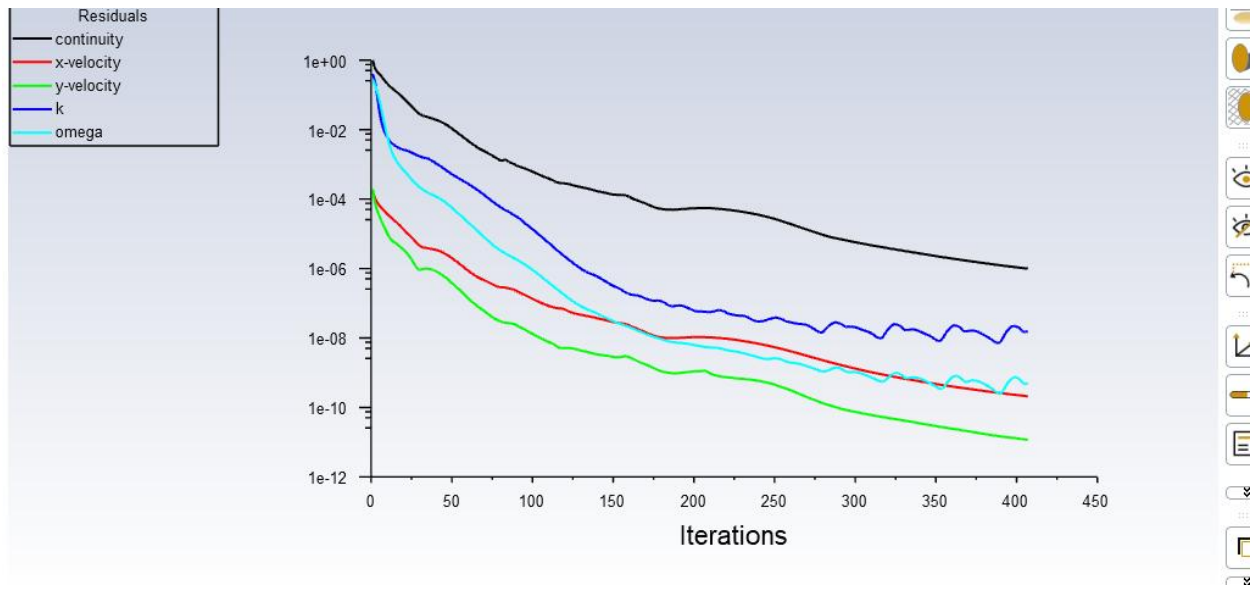
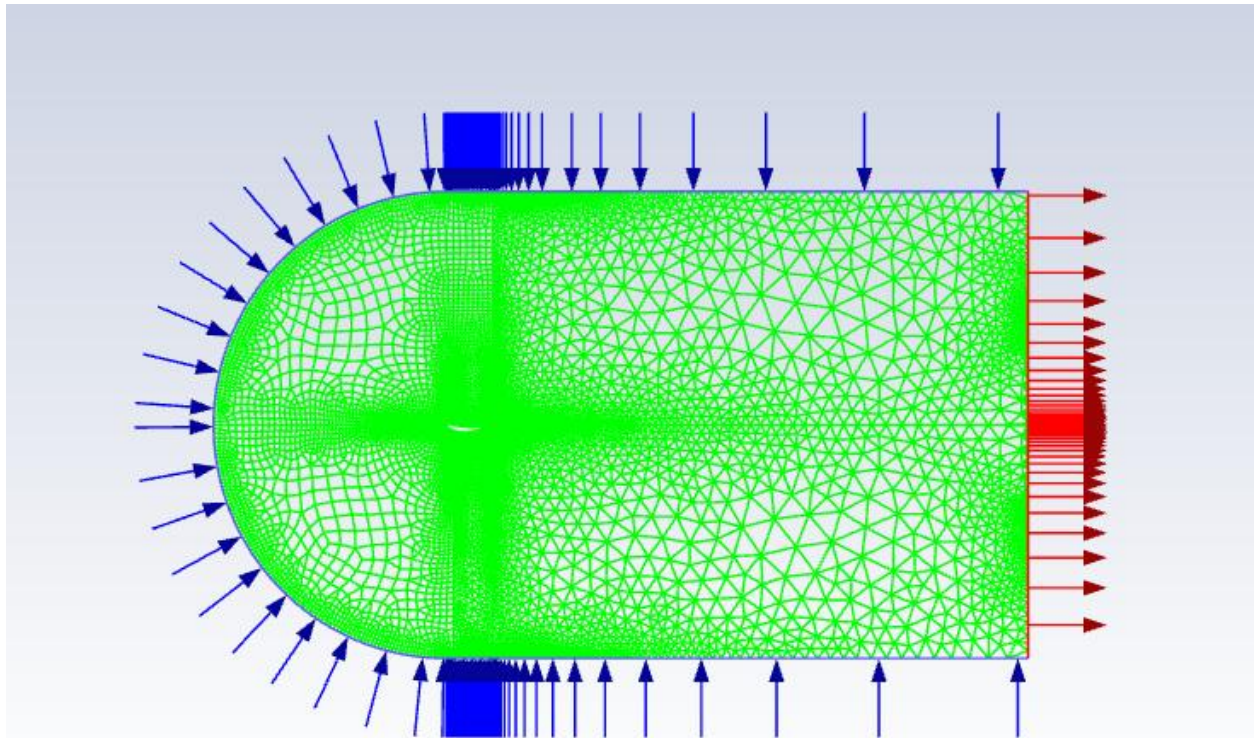


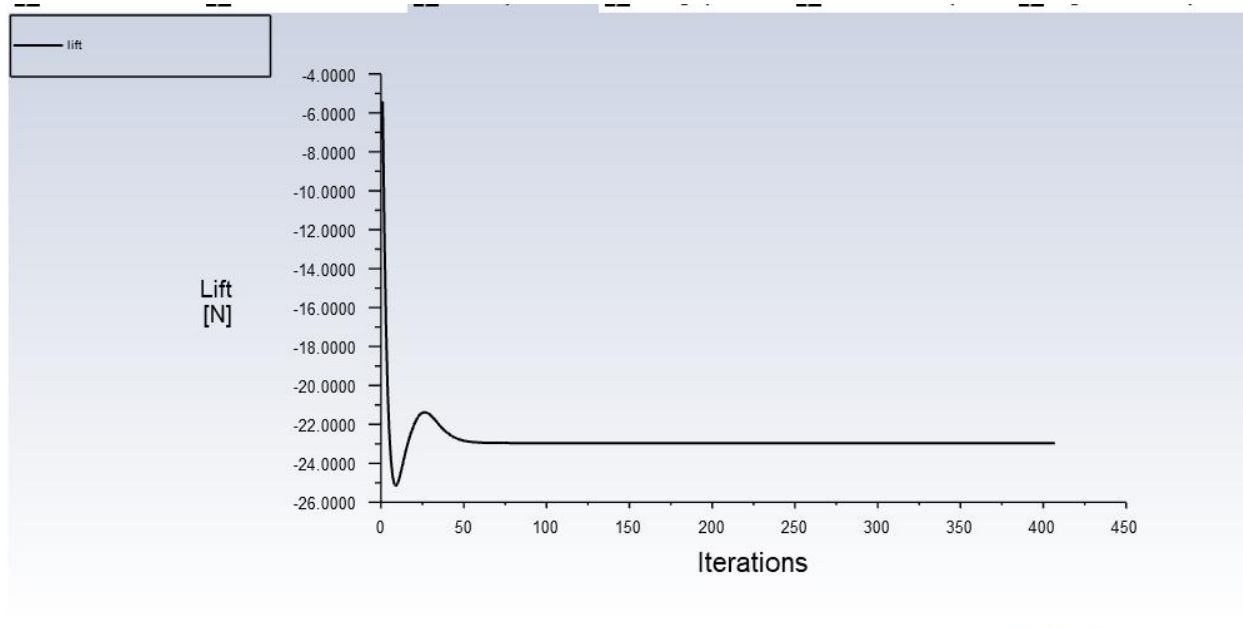
Quad4

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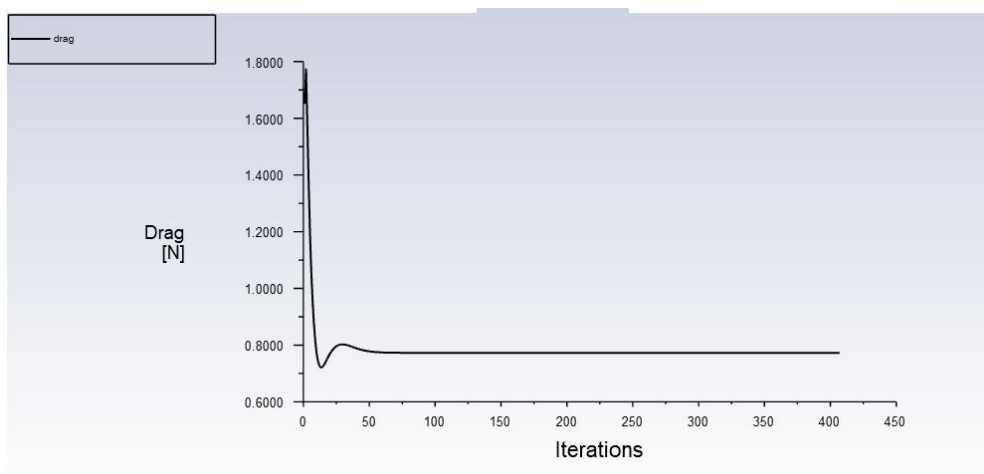




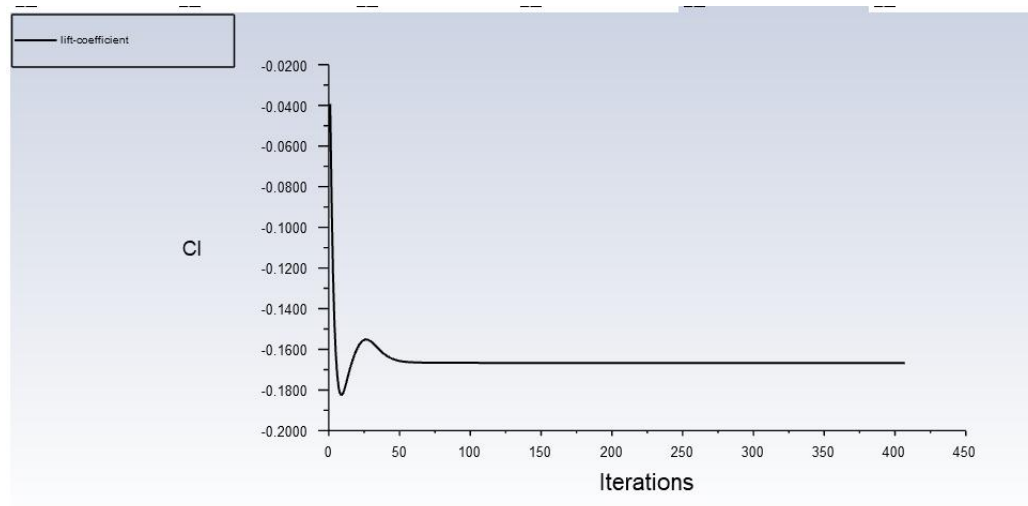


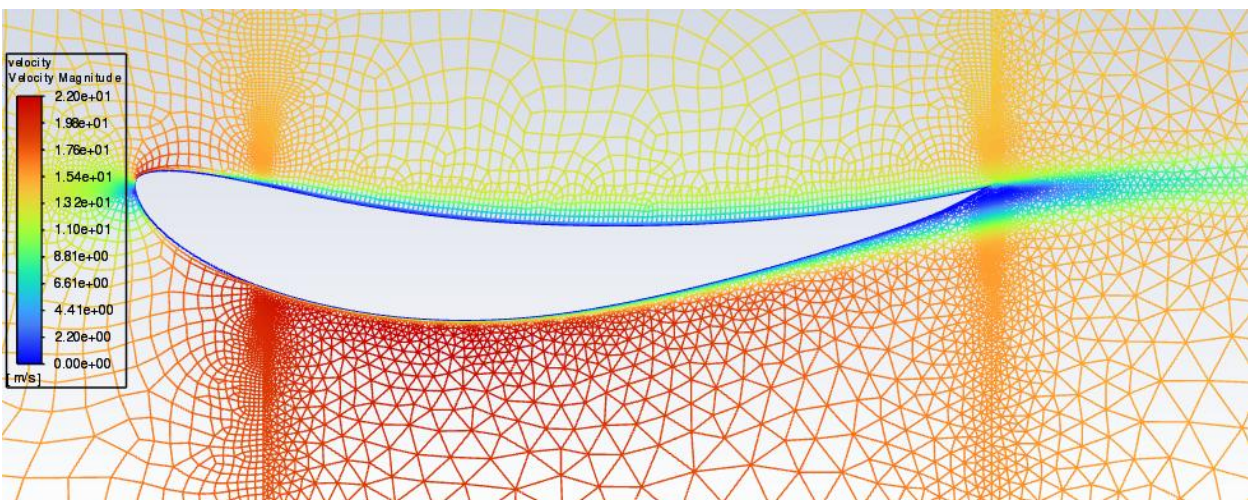
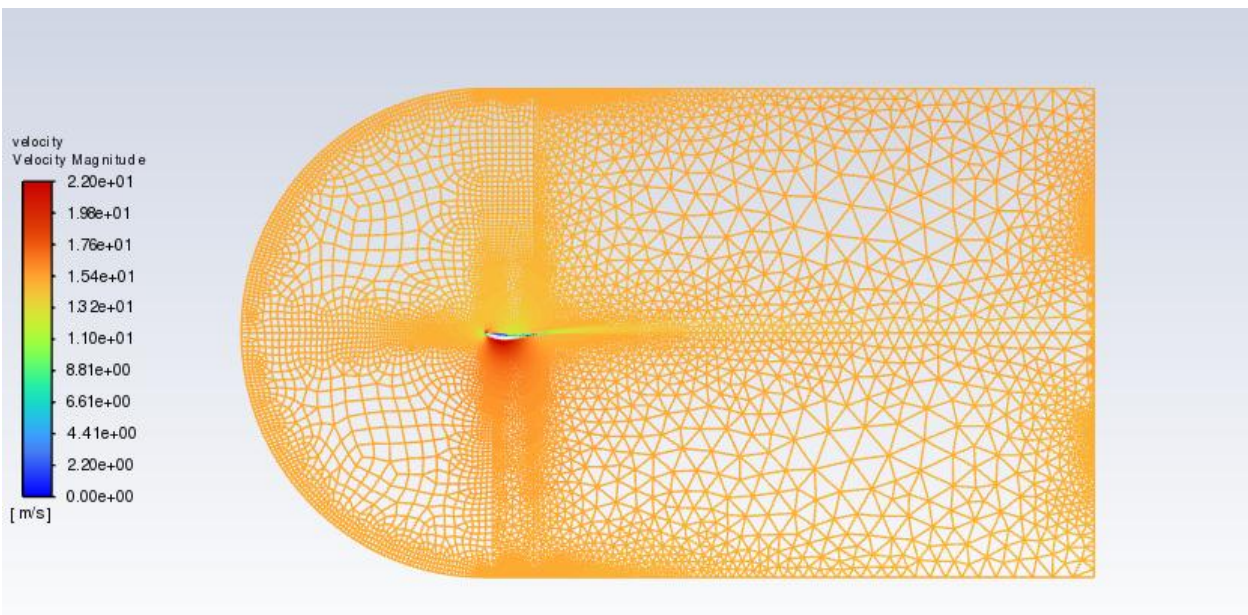
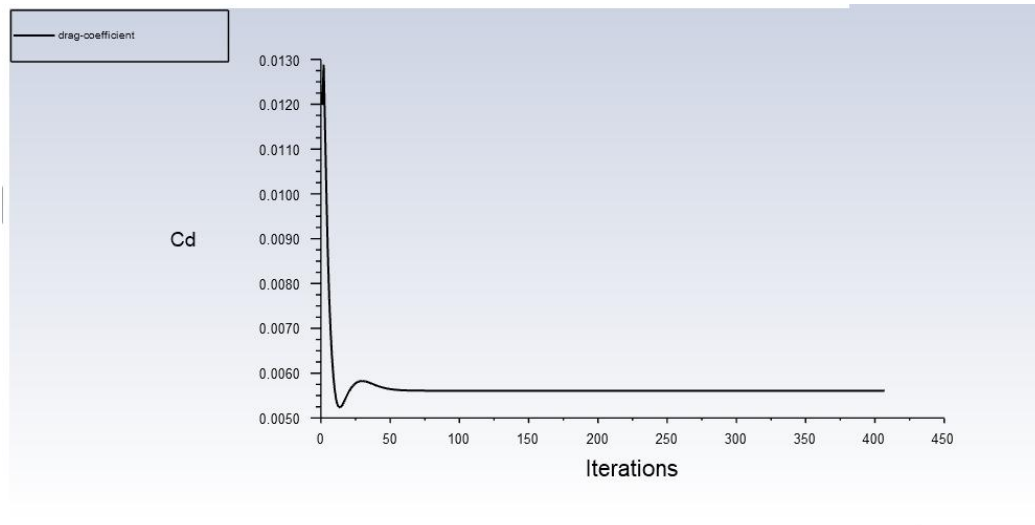


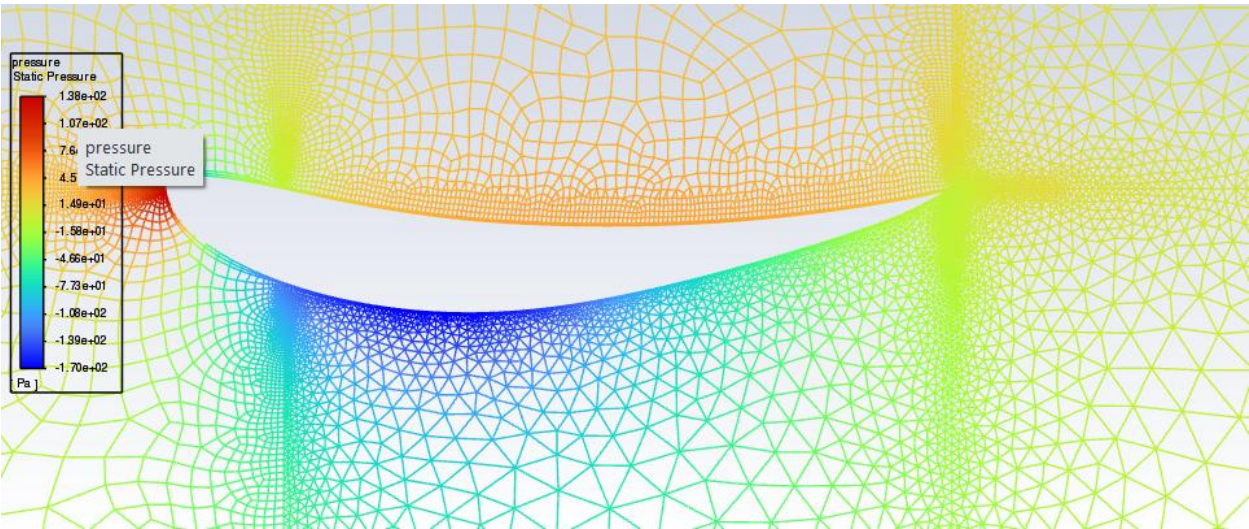
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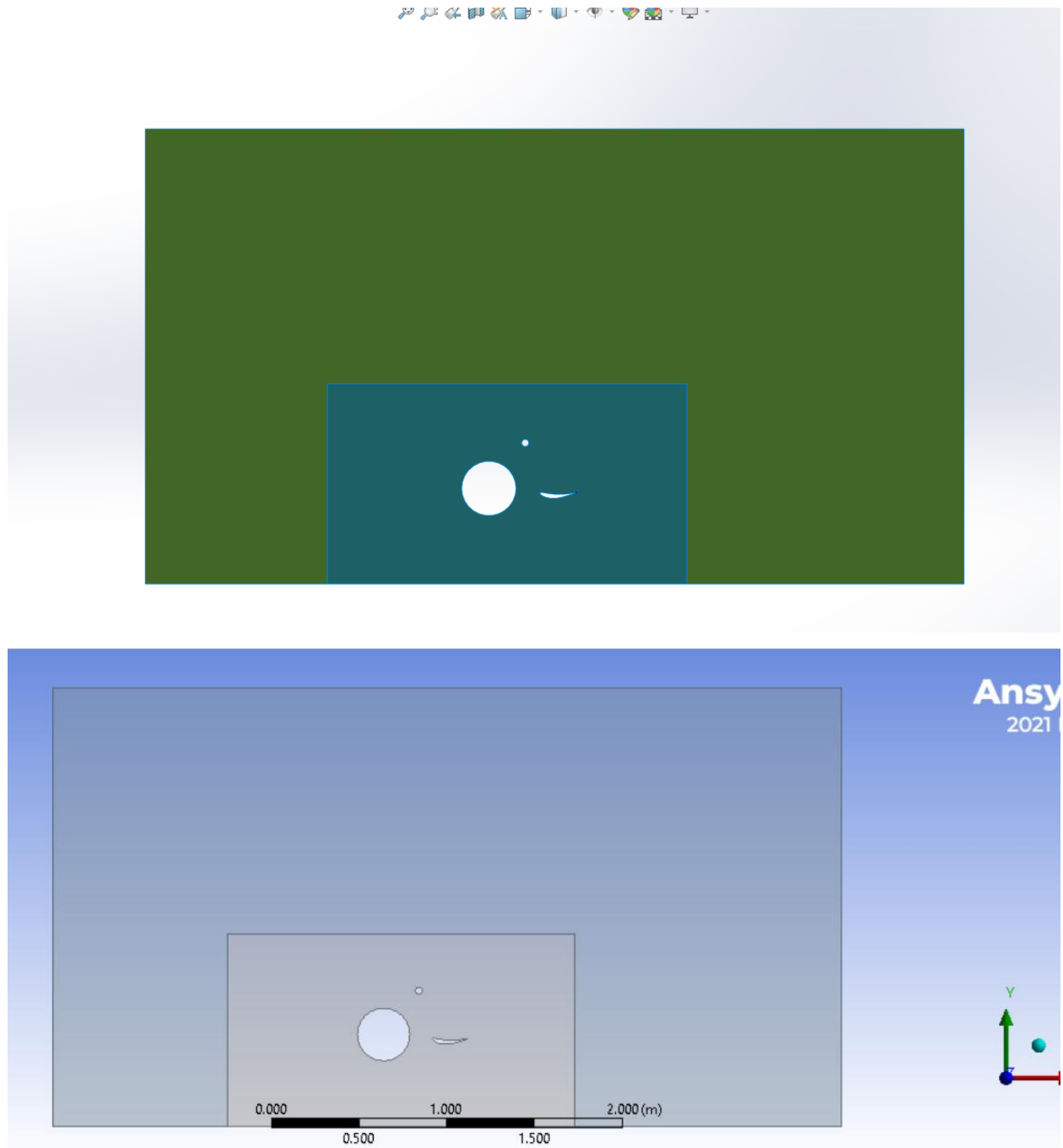


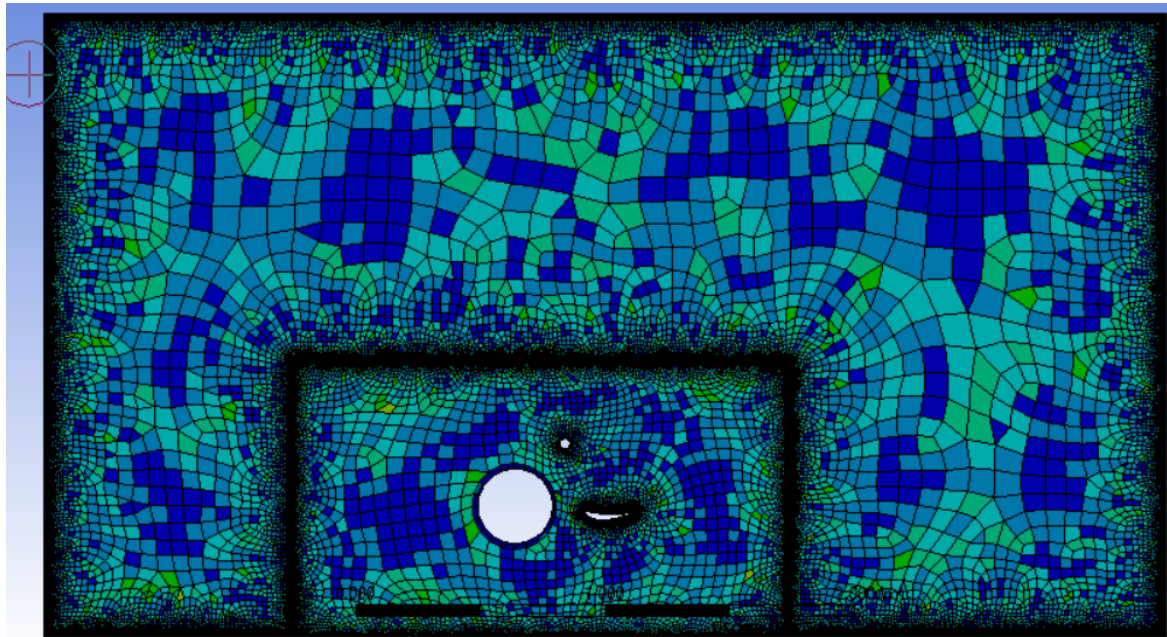
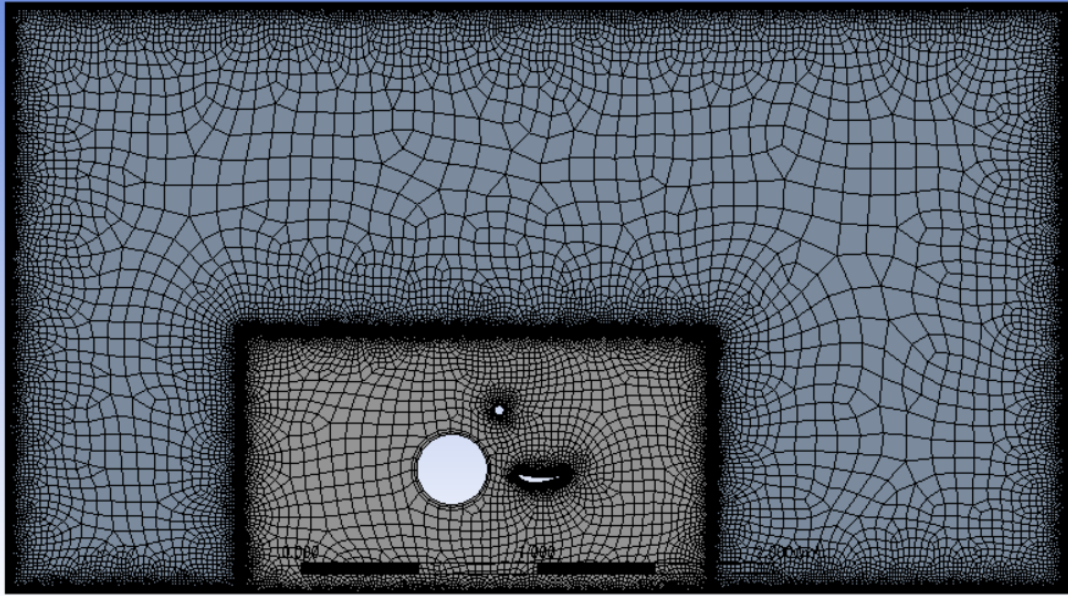


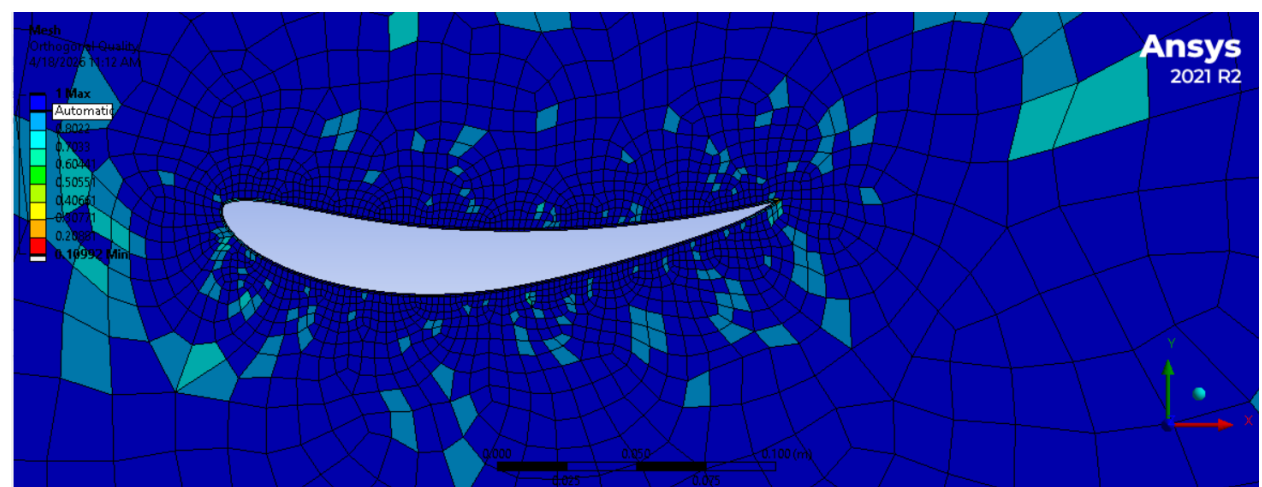
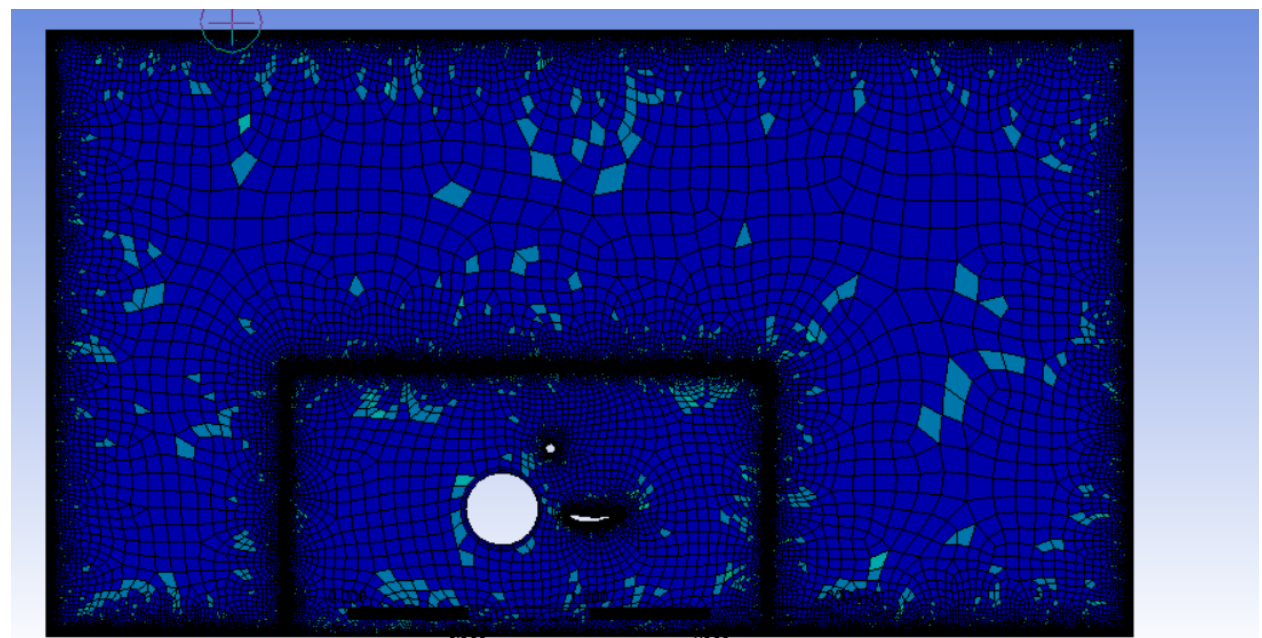
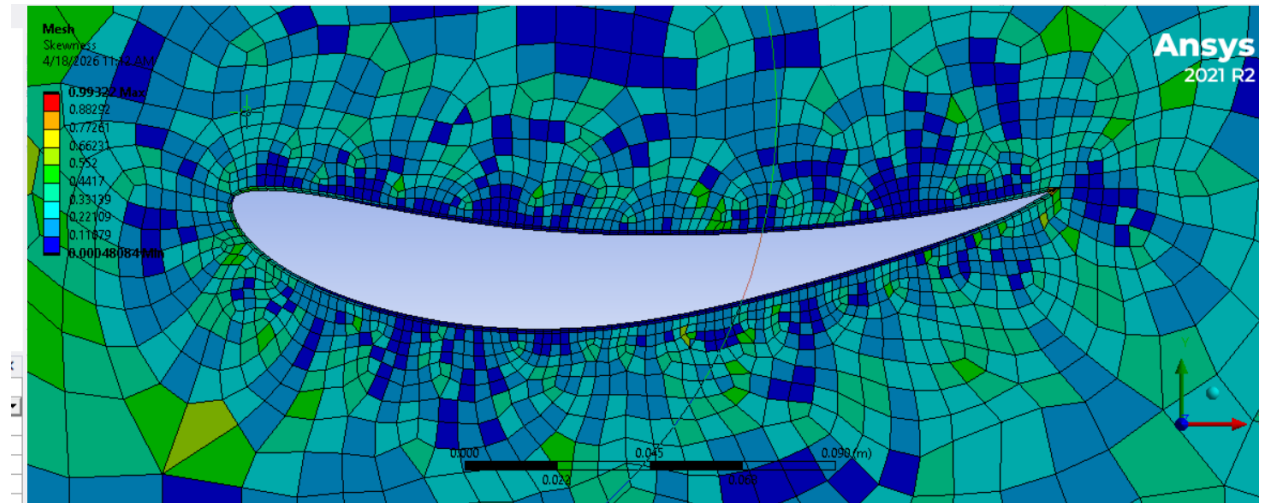
Results

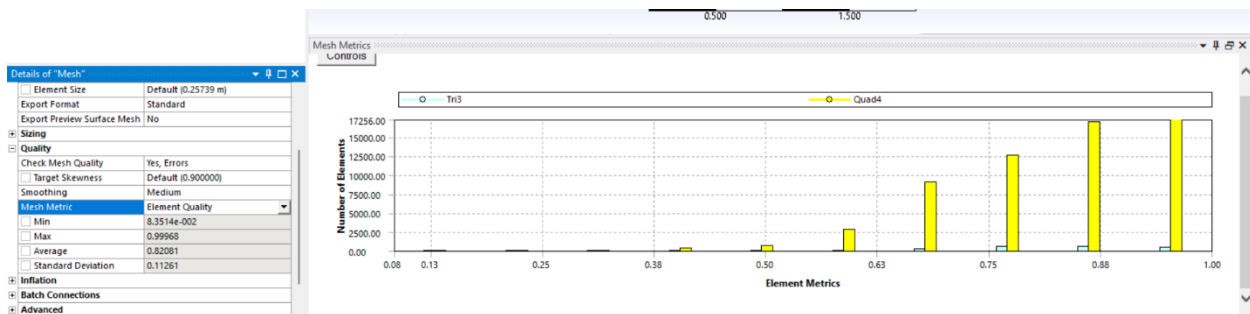
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#															
# The following header line defines the name of the columns by reference to the parameters.															
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DP 0	250	5000	150	300	150	100	0	200	1000	2500	2000	170	-22.9688	-0.16667	0.772771
DP 1	300	5000	200	400	200	150	0	200	1000	2500	2000	170	-22.316	-0.16193	0.795333
DP 2	310	5000	210	420	210	160	0	200	1000	2500	2000	170	-22.6919	-0.16466	0.784996
DP 3	320	5000	220	440	220	170	0	200	1000	2500	2000	170	-22.8013	-0.16545	0.790667
DP 4	330	5000	230	460	230	180	0	200	1000	2500	2000	170	-23.1781	-0.16819	0.782047
DP 5	340	5000	240	480	240	190	0	200	1000	2500	2000	170	-22.7351	-0.16497	0.793001
DP 6	250	5000	250	500	250	200	0	200	1000	2500	2000	170	-22.9616	-0.16662	0.782043
DP 7	350	5000	260	520	260	210	0	200	1000	2500	2000	170	-23.1645	-0.16809	0.781106
DP 8	360	5000	270	540	270	220	0	200	1000	2500	2000	170	-23.1238	-0.16779	0.780325
DP 9	370	5000	280	560	280	230	0	200	1000	2500	2000	170	-23.328	-0.16927	0.77902
DP 10	250	5000	150	300	150	100	0	200	1000	2500	2000	170	-22.9688	-0.16667	0.772771

Stimulation for E423 with changing chord length and AoA, with head and main hoop

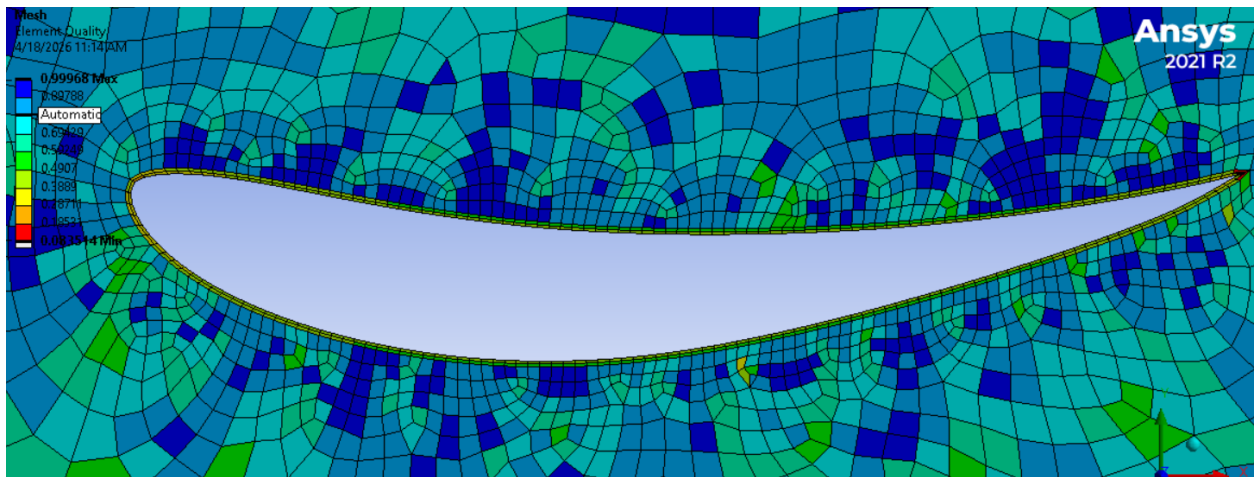
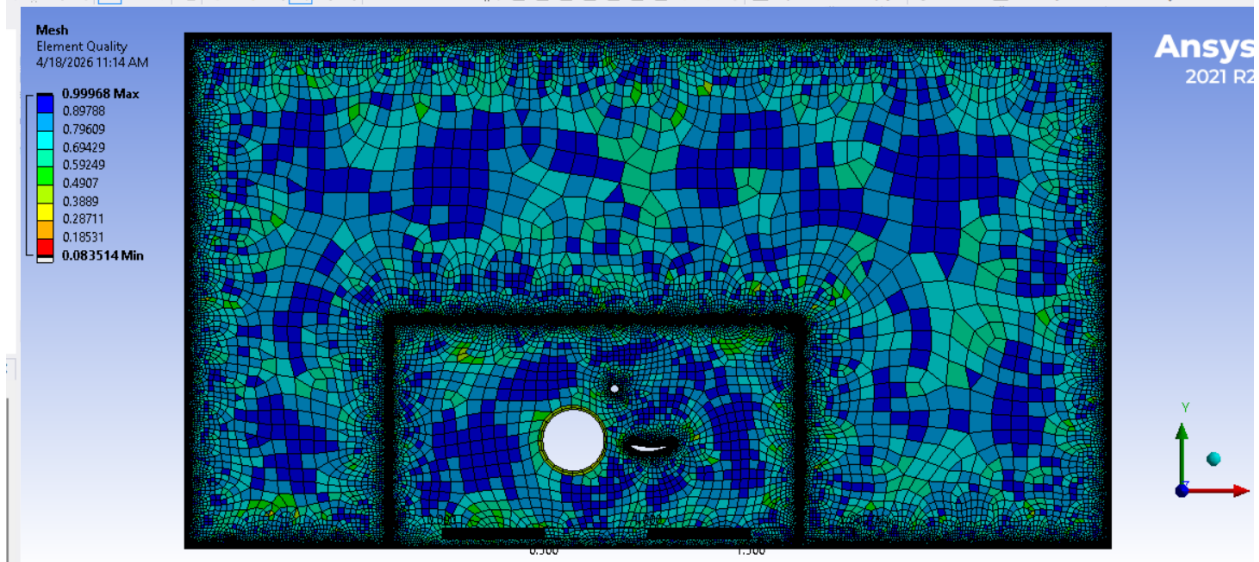
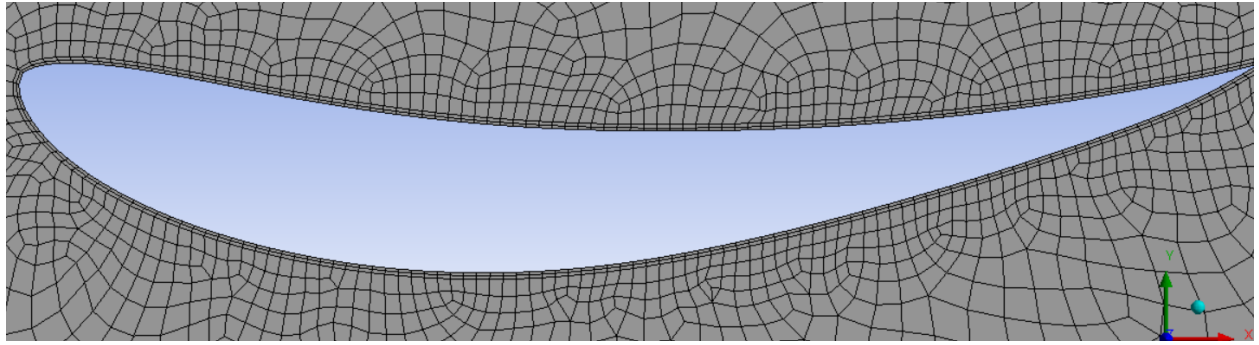


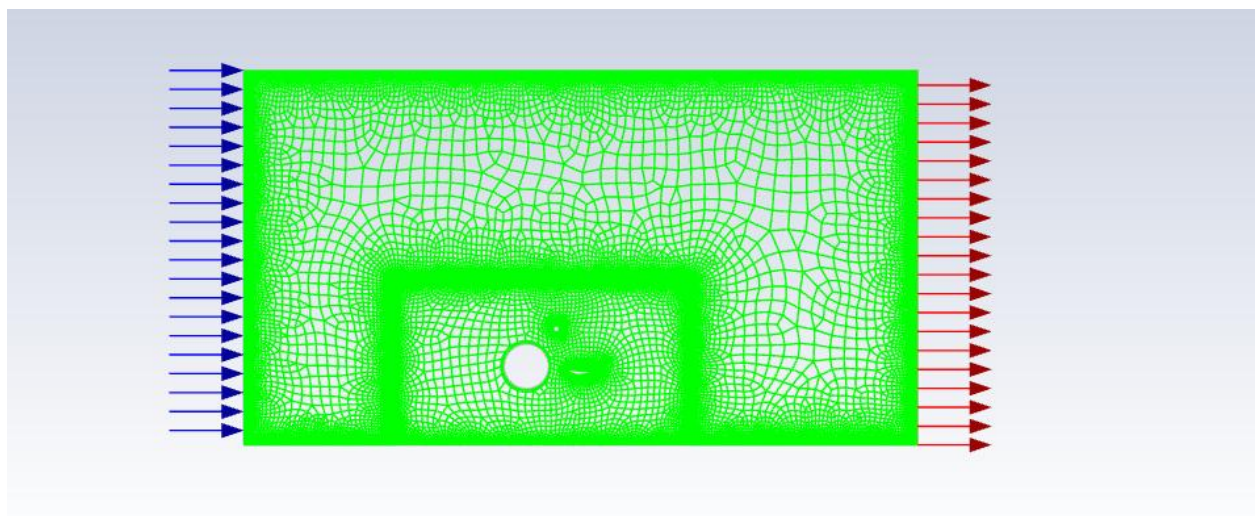
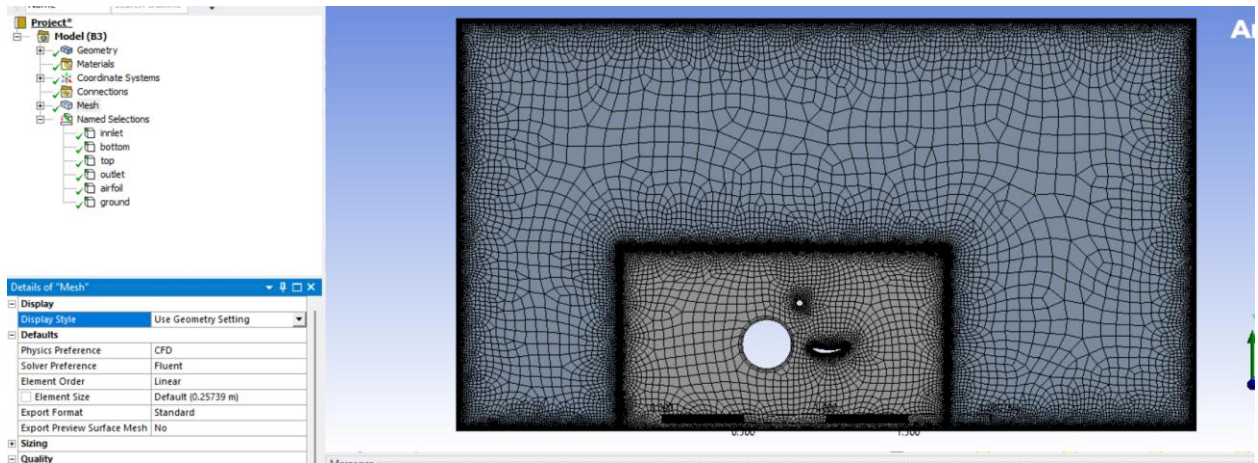


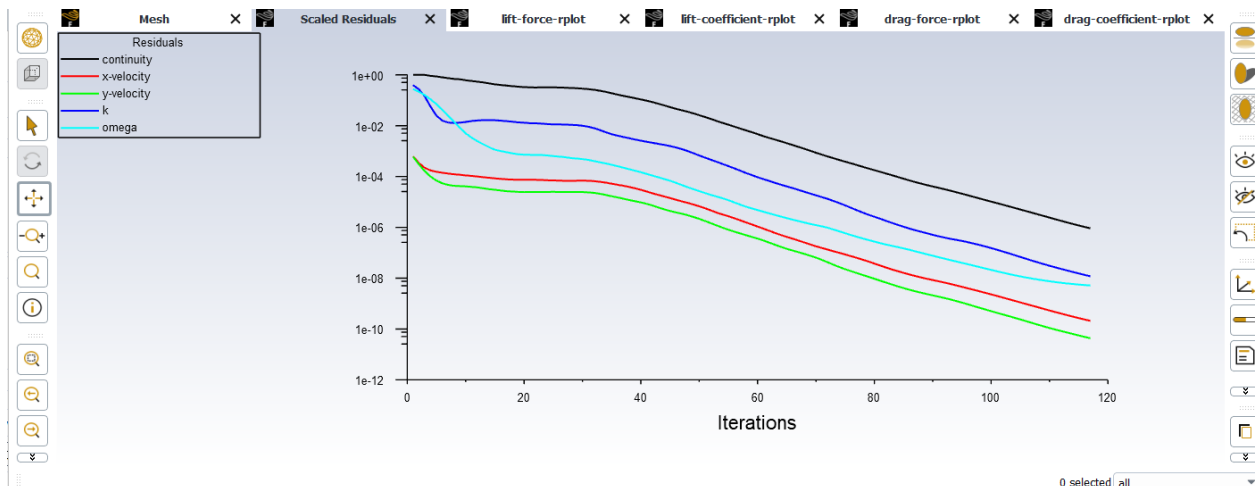




Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
☐ Transition Ratio	0.272
☐ Maximum Layers	2
☐ Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Batch Connections	
Advanced	
Statistics	

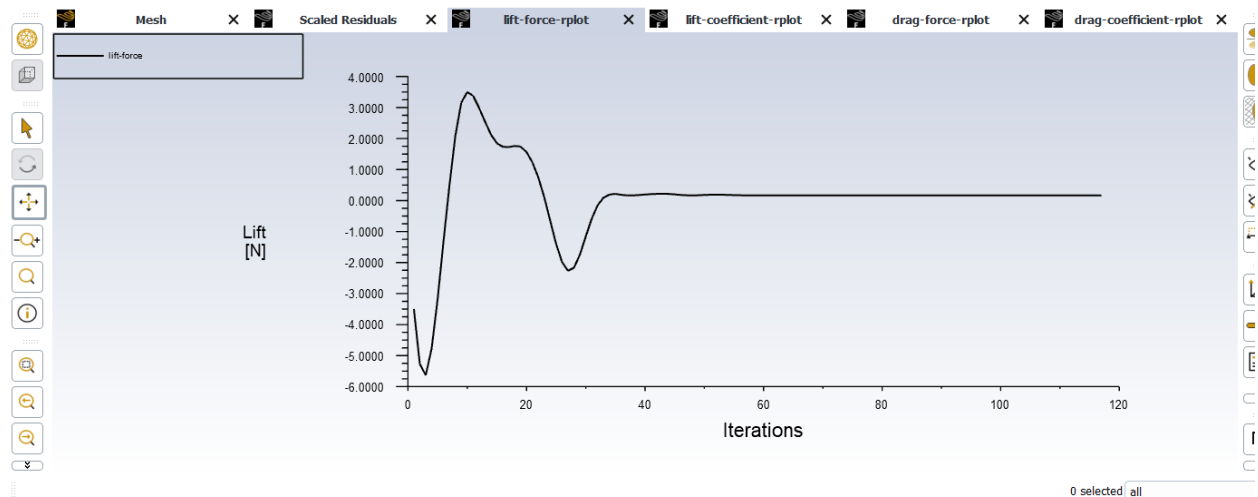






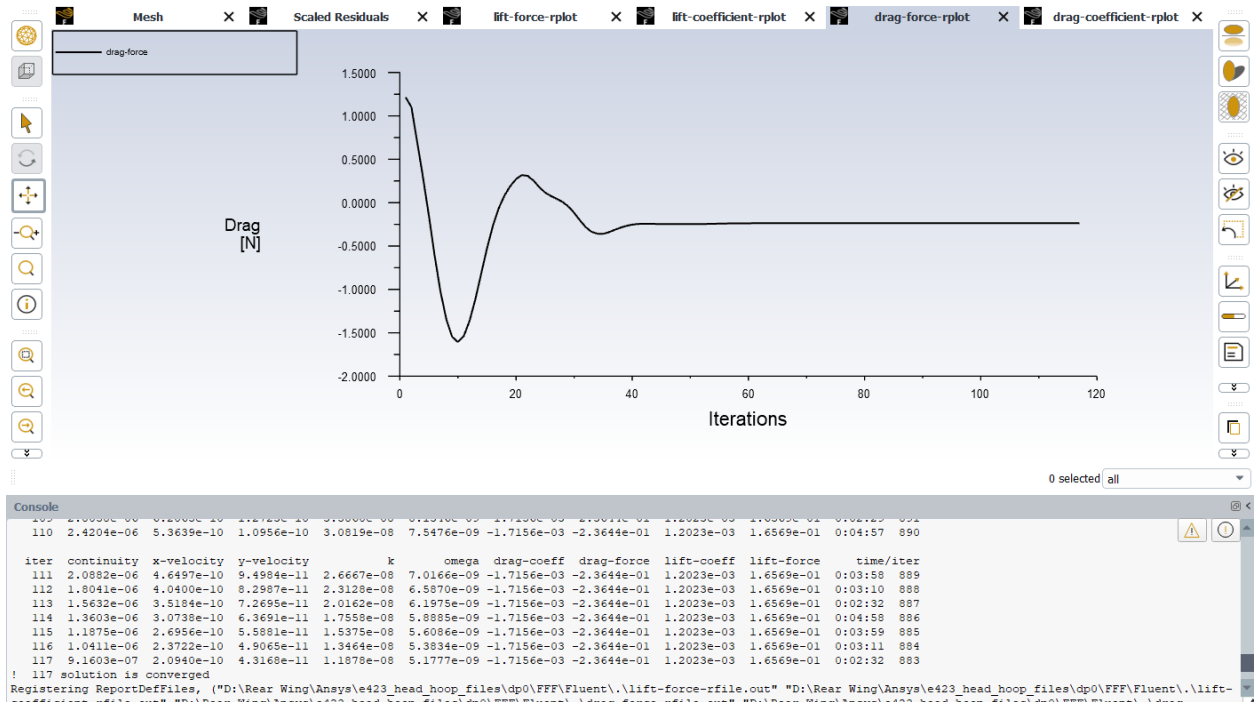
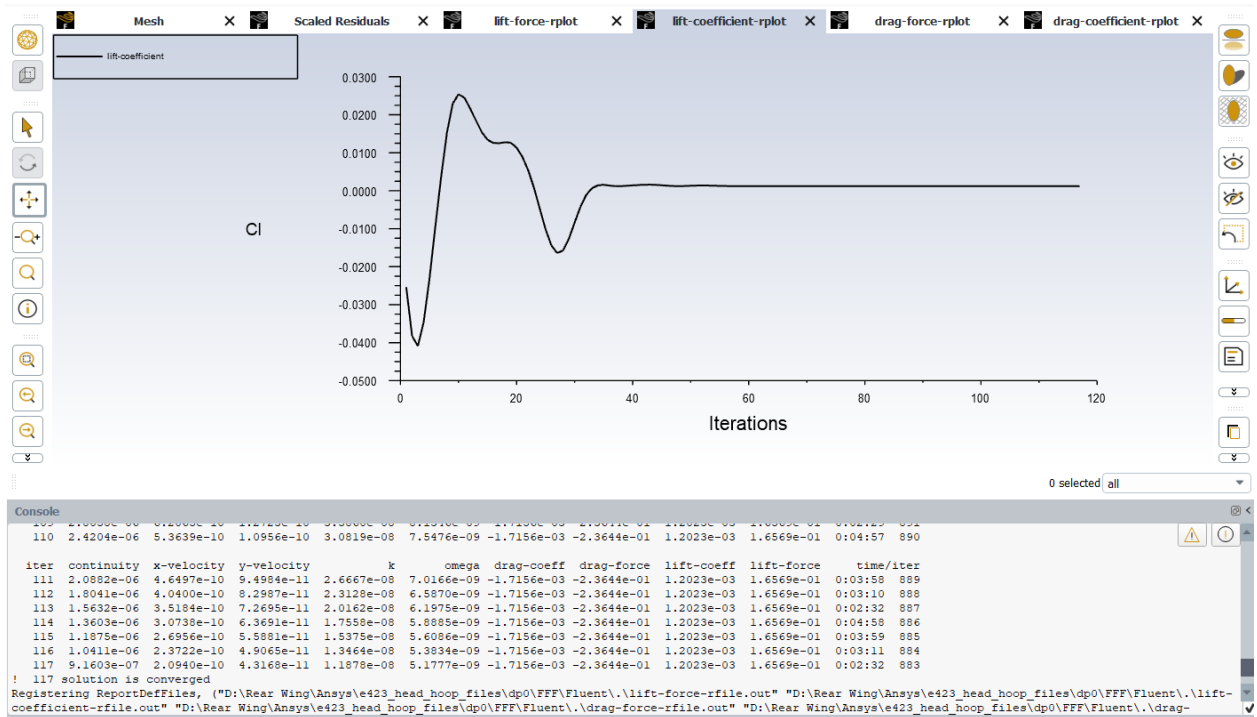
Console

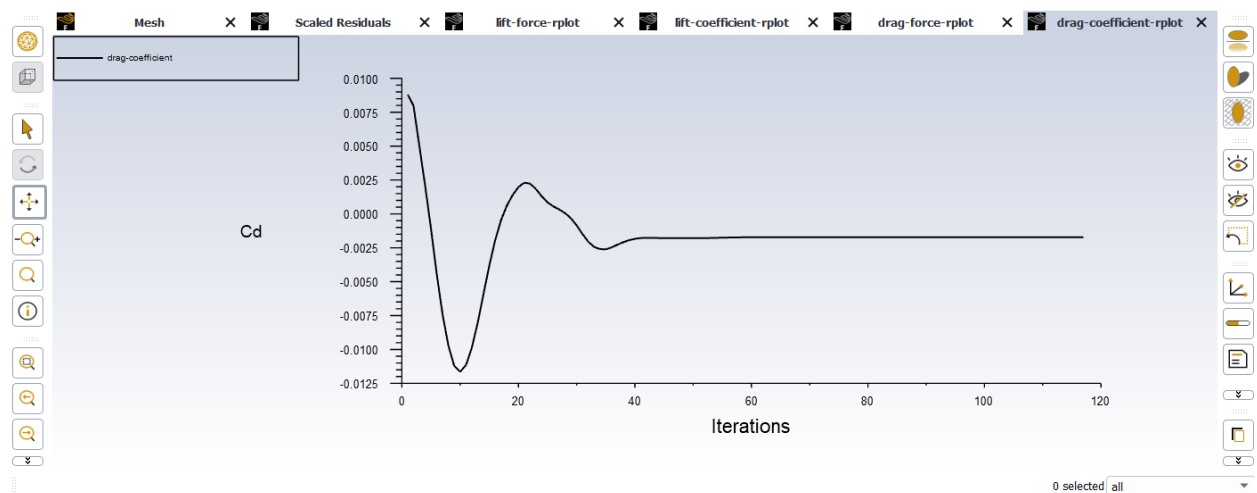
```
110 2.4204e-06 5.3639e-10 1.0956e-10 3.0819e-08 7.5476e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:04:57 890
iter continuity x-velocity y-velocity k omega drag-coeff drag-force lift-coeff lift-force time/iter
111 2.0882e-06 4.6497e-10 9.4984e-11 2.6667e-08 7.0166e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:58 889
112 1.8041e-06 4.0400e-10 8.2987e-11 2.3128e-08 6.5870e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:10 888
113 1.5632e-06 3.5184e-10 7.2695e-11 2.0162e-08 6.1975e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:02:32 887
114 1.3603e-06 3.0738e-10 6.3691e-11 1.7558e-08 5.8885e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:04:58 886
115 1.1875e-06 2.6956e-10 5.5881e-11 1.5375e-08 5.6086e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:59 885
116 1.0411e-06 2.3722e-10 4.9065e-11 1.3464e-08 5.3834e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:11 884
117 9.1603e-07 2.0940e-10 4.3168e-11 1.1878e-08 5.1777e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:02:32 883
! 117 solution is converged
Registering ReportDefFiles, ("D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\lift-force-rfile.out" "D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\lift-coefficient-rfile.out" "D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\drag-force-rfile.out" "D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\drag-coefficient-rfile.out")
```



Console

```
110 2.4204e-06 5.3639e-10 1.0956e-10 3.0819e-08 7.5476e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:04:57 890
iter continuity x-velocity y-velocity k omega drag-coeff drag-force lift-coeff lift-force time/iter
111 2.0882e-06 4.6497e-10 9.4984e-11 2.6667e-08 7.0166e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:58 889
112 1.8041e-06 4.0400e-10 8.2987e-11 2.3128e-08 6.5870e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:10 888
113 1.5632e-06 3.5184e-10 7.2695e-11 2.0162e-08 6.1975e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:02:32 887
114 1.3603e-06 3.0738e-10 6.3691e-11 1.7558e-08 5.8885e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:04:58 886
115 1.1875e-06 2.6956e-10 5.5881e-11 1.5375e-08 5.6086e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:59 885
116 1.0411e-06 2.3722e-10 4.9065e-11 1.3464e-08 5.3834e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:11 884
117 9.1603e-07 2.0940e-10 4.3168e-11 1.1878e-08 5.1777e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:02:32 883
! 117 solution is converged
Registering ReportDefFiles, ("D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\lift-force-rfile.out" "D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\lift-coefficient-rfile.out" "D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\drag-force-rfile.out" "D:\Rear Wing\Ansys\ae423_head_hoop_files\dp0\FFF\Fluent\.\drag-coefficient-rfile.out")
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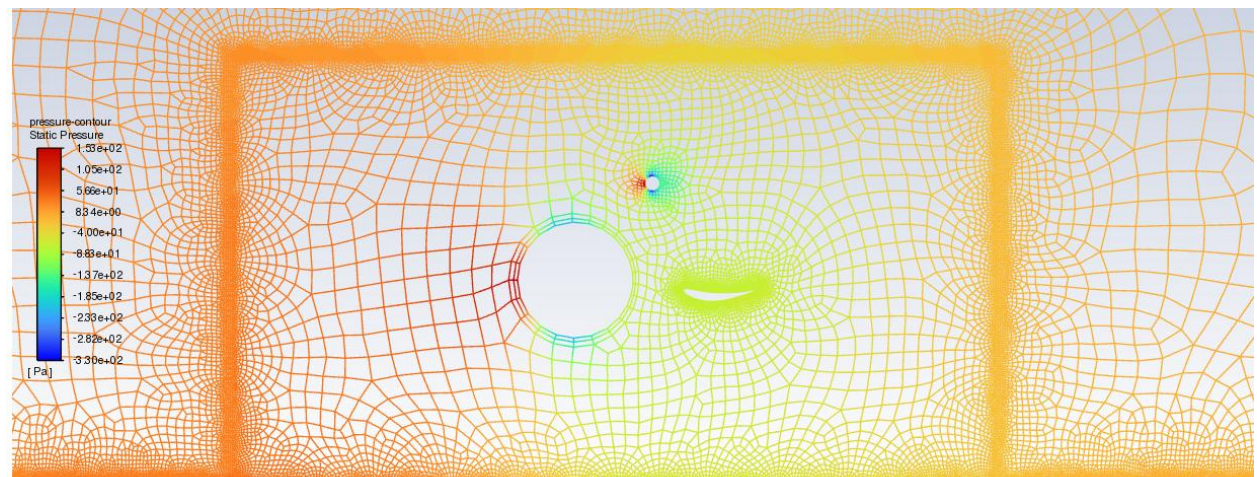


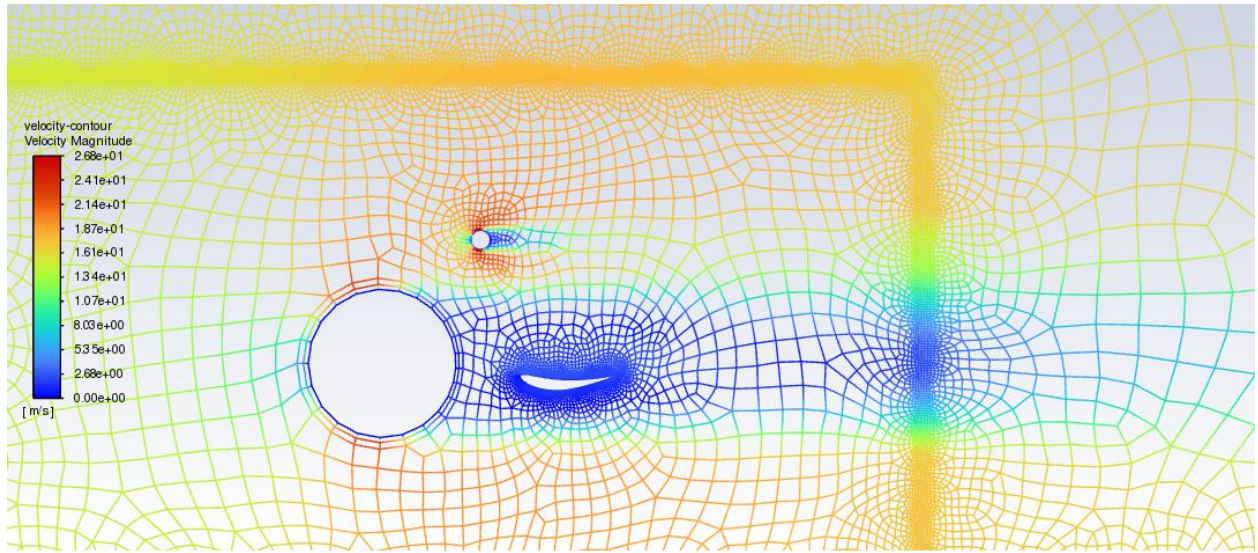
Console

```
109 2.7000e-06 5.1200e-10 1.1275e-10 3.7000e-06 7.1297e-09 1.7156e-03 1.7007e-01 1.2023e-03 1.6569e-01 0.02125 891
110 2.4204e-06 5.3639e-10 1.0956e-10 3.0819e-08 7.5476e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0.04:57 890

iter continuity x-velocity y-velocity k omega drag-coeff drag-force lift-coeff lift-force time/iter
111 2.0882e-06 4.6497e-10 9.4984e-11 2.6667e-08 7.0166e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:58 889
112 1.8041e-06 4.0400e-10 8.2987e-11 2.3128e-08 6.5870e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:10 888
113 1.5632e-06 3.5184e-10 7.2695e-11 2.0162e-08 6.1975e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:02:32 887
114 1.3603e-06 3.0738e-10 6.3691e-11 1.7558e-08 5.8885e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:04:58 886
115 1.1875e-06 2.6956e-10 5.5881e-11 1.5375e-08 5.6086e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:59 885
116 1.0411e-06 2.3722e-10 4.9065e-11 1.3464e-08 5.3834e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:03:11 884
117 9.1603e-07 2.0940e-10 4.3168e-11 1.1878e-08 5.1777e-09 -1.7156e-03 -2.3644e-01 1.2023e-03 1.6569e-01 0:02:32 883

! 117 solution is converged
Registering ReportDefFiles, ("D:\Rear Wing\Ansys\423 head hoop_files\dp0\FFF\Fluent\.\lift-force-rfile.out" "D:\Rear Wing\Ansys\423 head hoop_files\dp0\FFF\Fluent\.\lift-coefficient-rfile.out" "D:\Rear Wing\Ansys\423 head hoop_files\dp0\FFF\Fluent\.\drag-force-rfile.out" "D:\Rear Wing\Ansys\423 head hoop_files\dp0\FFF\Fluent\.\drag-
```





Results

W	X	Y	Z	AA	AB	AC
P23 - lift-force-op	P24 - lift-coefficient-op	P25 - drag-force-op	P26 - drag-coefficient-op	☐ Ret...	Retained Data	Note
N		N				
0.16569	0.0012023	-0.23644	-0.0017156	☒	✓	
0.27939	0.0020273	-0.23929	-0.0017363	☐		
0.55034	0.0039934	-0.30435	-0.0022084	☐		
0.51249	0.0037188	-0.32659	-0.0023698	☐		
0.49885	0.0036198	-0.30031	-0.0021791	☐		
0.5435	0.0039437	-0.36763	-0.0026676	☐		
⚡	⚡	⚡	⚡	☐		
⚡	⚡	⚡	⚡	☐		
⚡	⚡	⚡	⚡	☐		
⚡	⚡	⚡	⚡	☐		
⚡	⚡	⚡	⚡	☐		
⚡	⚡	⚡	⚡	☐		

P	Q	R	S	T
P16 - Edge Sizing Number of Divisions	P18 - Edge Sizing 3 Number of Divisions	P19 - Edge Sizing 2 Number of Divisions	P20 - Edge Sizing 4 Number of Divisions	P21 - velocity
				m s ⁻¹
250	1000	600	200	15
250	1000	600	200	15
1000	1500	2000	500	15
1010	1510	2010	510	15
1012	1512	2012	512	15
1014	1514	2015	515	15
1015	1520	2020	518	15
1020	1525	2022	520	15
1022	1527	2025	525	15
1024	1530	2030	530	15
1025	1535	2035	532	15
1028	1540	2040	535	15

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